

A Synopsis of Elementary Causality in Pure Holonic Mathematics

dependency edition 1 · exact constructions, transformations, derivations, boundaries

Truth boundary. Entries marked project-postulate define the laboratory modeling language. proved-standard imports ordinary mathematics under its stated hypotheses. proved-derived is a laboratory derivation. conditional retains its antecedent. conjecture is not proved. No entry in this edition proves the Riemann Hypothesis.

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1 Elementary situated relations

1.0.1 H.0001 Ambient mathematical foundation

Boundary · definition

statement The synopsis is interpreted in ordinary dependent type theory and standard mathematics. A **type** declares admissible terms; a proposition is a type whose inhabitants are proofs; equality, functions, relations, quotients, categories, topology, measure, and analysis retain their ordinary meanings.

boundary Holonics is not presently a proved replacement foundation. A claim that all mathematics admits a conservative holonic translation remains a conjectural programme until a translation and conservativity theorem are supplied.

source Laboratory regrade; formal carrier: Lean 4 with Mathlib where mechanized.

1.0.2 H.0002 Situated admissibility

Postulate · project-postulate

depends: H.0001

statement An admissible datum is never introduced as an untyped bare glyph. It is supplied as

$$x : X_{\theta,e}$$

in a declared type or fiber, parameter θ , event cut e , source boundary, and incidence context.

transformations Recharting may change the presentation of x only through an explicitly admitted map between the relevant fibers.

boundary This is a modeling discipline, not a denial that ordinary mathematics may quantify over elements after their ambient structure has been fixed.

source Brandon's first-axiom discussions, regraded as a project postulate rather than an axiom of all mathematics.

1.0.3 H.0003 Participating receiver

Postulate · project-postulate

depends: H.0002

statement Every comparison or observation is made through a declared receiver

$$\rho : X \rightarrow Y$$

or receiver relation $\rho \subseteq X \times Y$. The observed face is $\rho(x)$ or the fiber $\rho(x, -)$; the receiver and its parameterization are part of the complete comparison.

receiver A receiver need not be an agent or a camera. Evaluation, projection, integration, quotient, spectrum, trace, compiler, and physical instrument are all possible receiver species.

boundary Receiver dependence does not make every statement arbitrary. Naturality, invariance, and conservative transport can relate different receivers exactly.

source Laboratory observer/first-person synthesis; categorical carrier supplied below rather than assumed here.

1.0.4 H.0004 Causal attribution

Postulate · project-postulate

depends: H.0002, H.0003

statement A transformation retains every source, incidence, orientation, and path distinction required by its declared future receiver family. Material may depart only after the retained factor still determines those future faces.

boundary This does not require a perfect copy of every event. It constrains only the factorization claimed to remain exact.

source Laboratory lineage/compression correction.

1.0.5 H.0005 Exact construction precedes numerical presentation

Boundary · project-postulate

depends: H.0001, H.0003, H.0004

statement The mathematical carrier is retained exactly: as a typed ratio, algebraic element, symbolic expression, formal series, convergent series with its topology, exact interval with a certificate, or another construction whose equality law is declared. Proof-bearing transformations operate on these exact carriers.

transformations Closure, convergence, density, and asymptotic statements remain exact when written with their full quantifiers. For example,

$$x \in \overline{S} \leftrightarrow \forall U \text{ open} : x \in U \rightarrow U \cap S \neq \emptyset.$$

No sampled agreement or finite surrogate is introduced by this definition.

boundary This discipline does not reject exact real or complex numbers, limits, measure, topology, or analysis. A finite display is not a proof-bearing replacement for the construction it receives.

source src/holobrochos/CANON/06_THE_PURE_BIT.md, regraded from historical implementation bans into an exact mathematical evidentiary boundary.

1.0.6 H.0006 Current and form are receiver-relative roles

Definition · definition

depends: H.0002, H.0003, H.0004

statement In a situated transition $x_t \rightarrow x_{t+1}$, **current** names the relation while it is being conducted through the selected cut; **form** names a consequential factor of that relation received as standing for a later cut. The same carrier can be current for one receiver and form for another, and the roles may exchange under a later transition.

transformations Depositing form is not storing the complete live current. It is a receiver-exact factorization of consequence; remounting that factor in a later event creates a new situated current.

boundary Current and form are not declared as disjoint ontological types. The stronger claim that all physical substance is literally one current remains a laboratory hypothesis, not a theorem imported here.

source src/holobrochos/INTUITIONS/lineages/the-current-and-the-form.md, with its roles-not-kinds correction retained.

1.0.7 H.0007 Response and path receiver

Definition · definition

depends: H.0003, H.0004, H.0006

statement A path receiver presents a composable transition lineage

$$x_0 \rightarrow x_1 \rightarrow \cdots \rightarrow x_n$$

together with selected differences between successive faces. A response is the changed receiver relation induced by contact, not a scalar score attached to a lone occurrence.

receiver Endpoint evaluation, the complete trace, a boundary flux, a winding, and a residual are different receivers of one path. None is silently crowned as its complete identity.

boundary “Recognition is navigation” is the laboratory interpretation motivating this carrier. It does not assert that every biological or computational recognition process is already represented by one chosen path category.

source src/holobrochos/INTUITIONS/lineages/recognition-navigation-bundle.md, regraded into a precise receiver definition.

1.0.8 H.0010 Typed relation and correspondence

Definition · definition

depends: H.0001

statement A relation from X to Y is a predicate

$$R : X \times Y \rightarrow \text{Prop}.$$

Its converse is $R^{-1}(y, x) \leftrightarrow R(x, y)$. The composite of $R \subseteq X \times Y$ and $S \subseteq Y \times Z$ is

$$(S \circ R)(x, z) \leftrightarrow \exists y \in Y : R(x, y) \wedge S(y, z).$$

A function is a total single-valued relation.

derivation Associativity follows by reassociating the two existential witnesses:

$$\exists y, z : R(x, y) \wedge S(y, z) \wedge T(z, w).$$

Identity is equality on each type.

boundary Composition records admissible endpoint incidence. It does not by itself retain the interior witness y ; a path construction below does.

source Standard relation theory.

1.0.9 H.0011 Situated occurrence

Definition · definition

depends: H.0002, H.0010

statement A situated occurrence is a term $x : X_{\theta, e}$ together with the supplied source mark, incidence, orientation, and admitted transition relations at (θ, e) . Two occurrences may have equal receiver values while remaining distinct terms or path positions.

boundary Equality of coordinates, labels, serialized bytes, or hashes is not occurrence identity unless the relevant map is proved injective.

source Laboratory definition; refines the earlier reusable definition:situated-occurrence.

1.0.10 H.0012 Receiver face and fiber

Definition · definition

depends: H.0003, H.0010, H.0011

statement For a receiver $\rho : X \rightarrow Y$, the face of x is $\rho(x)$. For a relational receiver $\rho \subseteq X \times Y$, the face is the fiber

$$\rho_x = \{y \in Y : \rho(x, y)\}.$$

The receiver kernel

$$x \underset{\rho}{\sim} x' \leftrightarrow \rho(x) = \rho(x')$$

is an equivalence relation when ρ is functional.

derivation Reflexivity, symmetry, and transitivity are inherited from equality in Y .

receiver The quotient $\frac{X}{\underset{\rho}{\sim}}$ is the coarsest face retaining exactly one receiver and no claim about any other receiver.

boundary A face can be exact without reconstructing its complete causal interior.

source Standard kernel relation with laboratory receiver interpretation.

1.0.11 H.0013 Comparison cell and oriented residual

Definition · definition

depends: H.0010, H.0012

statement Let $f, g : X \rightarrow Y$ be parallel paths or maps and $\rho : Y \rightarrow Z$ a receiver. Their comparison cell at x is

$$\rho(f(x)) - \rho(g(x)).$$

In an additive target its oriented residual is

$$\Delta_{\rho; f, g}(x) = \rho(f(x)) - \rho(g(x)).$$

transformations RIDE: the residual is zero on already admitted support. FOUND: closing the comparison requires adjoining a new typed support or law. OPEN: neither closure has been established.

boundary OPEN is a complete status of the present comparison, not negation, empty information, or automatic permission to invent a filler.

source Laboratory comparison/open-square synthesis.

1.0.12 H.0014 Relative holon

Definition · definition

depends: H.0010, H.0011

statement Given a transition family T on X and an admitted boundary family B , a subobject $H \subseteq X$ is a holon relative to (T, B) when every T -transition whose participating boundary is internal to H closes in H . Transitions crossing B remain explicit exterior incidences.

boundary Closure is relative to (T, B) . No object is absolutely whole, and no relative holon is required to contain all contexts in which its members may participate.

source Laboratory definition, regraded into standard closure language.

1.0.13 H.0015 Soul and hierarchy of sameness

Definition · definition

depends: H.0011, H.0013, H.0014

statement A soul is a marked causal diagram: typed occurrences, source marks, incidence, admitted transitions, and receivers. Distinguish: Occurrence equality, diagram isomorphism, receiver equivalence, and equal serialization are distinct relations. Occurrence equality implies equal faces under every well-typed receiver by substitution; the converse generally fails. Every other implication requires its own conservative, injective, full, or faithful transport theorem.

transformations A lawful soul-equivalence is an isomorphism of the marked diagrams which commutes with every receiver named by the comparison.

boundary Yoneda identifies an object only through the complete natural family of maps into or out of it, up to isomorphism. One scalar, cross-ratio, image, or digest is not that family.

source src/holobrochos/CANON/03_THE_SEMANTICS.md, regraded through the standard Yoneda and typed-equality boundaries.

1.0.14 H.0016 Receiver-exact compression

Definition · definition

depends: H.0004, H.0012, H.0015

statement Let $\mathcal{R}$ be a declared family of future receivers on X . A compression is a quotient $q : X \rightarrow Q$ such that every $\rho \in \mathcal{R}$ factors:

$$\rho = \bar{\rho} \circ q.$$

The induced equivalence is

$$x \sim_{\mathcal{R}} y \leftrightarrow \forall \rho \in \mathcal{R} : \rho(x) = \rho(y).$$

derivation The canonical map to the quotient by $\sim_{\mathcal{R}}$ has the universal factorization property. Conversely, any common factor q identifies only pairs indistinguishable by every factored receiver.

transformations An inactive interior may depart when the active future receiver family factors through the emitted quotient. Enlarging $\mathcal{R}$ can refine or invalidate the compression.

boundary Compression is not synonymous with fewer bytes, deduplication, averaging, or a scalar quotient.

source Standard quotient factorization with laboratory compression interpretation.

1.0.15 H.0017 Standing and lineage

Definition · definition

depends: H.0011, H.0016

statement A lineage is a compatible family of consequential factors

$$q_{t(x_t)} \in Q_t$$

connected by transition maps $u_{t,t+1}$ such that

$$u_{t,t+1}(q_{t(x_t)}) = q_{t+1}(x_{t+1}).$$

Standing at cut t is the part of this compatible relation admitted as terrain for later transitions.

boundary Lineage is continuity of consequential relation, not retention of every constituent or a perfect ancestry crystal.

source Laboratory standing/lineage correction.

1.0.16 H.0018 Parameter family, phase, and seam

Definition · definition

depends: H.0010, H.0017

statement A parameterized law is a family T_θ over a parameter space Θ . A phase is a maximal connected stratum on which the declared law class, rank, orientation type, and continuation data remain locally equivalent. A seam is a component of the discriminant where at least one of those invariants changes.

transformations Quantization is a typed discrete invariant change at a seam. A phase transition need not be a change in spatial volume.

boundary The phase invariants must be named. “Different appearance” under one receiver is not by itself a phase transition.

source Laboratory phase correction in standard stratification language.

1.0.17 H.0019 Conditional recurrence field and loss

Definition · definition

depends: H.0012, H.0013, H.0018

statement For a receiver ρ , admitted history $\mathcal{H}$, and event family E , a probability face is a conditional measure or valuation

$$P_{\rho(E \mid \mathcal{H})}.$$

It records recurrence under the receiver’s analogy classes. A loss is an oriented residual $L_{\rho(f,g)}$ comparing two admitted transports or consequences.

transformations Bayes’ rule is the change of conditional receiver

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$$

whenever $P(B) \neq 0$. It changes the factorization of one joint measure; it does not introduce ontological chance.

boundary A probability field need not collapse to one scalar outside a declared event algebra. Loss has no intrinsic moral, reward, or punishment sign.

source Standard conditional probability with laboratory ontological regrade.

2 Logic, categories, gluing, and translation

2.0.1 H.0020 Typed equality and substitution

Rule · proved-standard

depends: H.0001

statement Equality $x \stackrel{\bar{x}}{=} y$ is an equivalence relation on terms of one type X . For every well-typed predicate or construction P on X , equality permits substitution of y for x . Equality across different ambient types requires an explicitly supplied transport.

transformations Construction equality implies equality under every function with source X . Equality under one noninjective receiver need not imply construction equality.

boundary A shared glyph, coordinate, unit label, or output value does not create a typed equality proof.

source Identity elimination in dependent type theory.

2.0.2 H.0021 Well-founded induction and recursion

Theorem · proved-standard

depends: H.0020, H.0010

statement If $\prec$ is well founded on X and

$$(\forall y \prec x : P(y)) \rightarrow P(x)$$

for every x , then $\forall x : P(x)$. Compatible predecessor data define a unique recursive function by the well-founded recursion theorem.

transformations Ordinary induction is the specialization to the predecessor relation on $\mathbb{N}$. Structural induction uses the immediate-subterm relation.

boundary A recurring visual pattern is not an induction hypothesis. The carrier, well-founded order, base cases, and successor implication must be stated.

source Standard well-founded induction and recursion theorem.

2.0.3 H.0022 Category and free path category

Definition · definition

depends: H.0010, H.0007

statement A category has typed objects and arrows, associative composition, and one identity arrow at each object. Every directed graph G generates a free category $\text{Path}(G)$ whose arrows are finite composable edge words and whose composition is concatenation.

receiver Endpoint evaluation forgets the interior word. The free path category retains it until a declared relation quotients paths.

boundary Category composition records lawful composability, not temporal causation by itself.

source Standard category theory and the free-category construction.

2.0.4 H.0023 Functor and natural transformation

Definition · definition

depends: H.0022

statement A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ preserves source, target, identities, and composition. A natural transformation $\eta : F \rightarrow G$ obeys

$$G(f) \circ \eta_X = \eta_Y \circ F(f)$$

for every $f : X \rightarrow Y$.

transformations Functorial receivers preserve execution composition. A lawful change of receiver is natural only when every such square commutes.

boundary A collection of objectwise translations is not a natural transformation until the arrow law is proved.

source Standard category theory.

2.0.5 H.0024 Universal property, limit, and colimit

Definition · definition

depends: H.0022, H.0023

statement A limit of $F : J \rightarrow \mathcal{C}$ is a terminal cone to F ; a colimit is an initial cocone from F . Either object is unique up to a unique isomorphism compatible with its structure maps.

transformations Products, equalizers, pullbacks, coproducts, coequalizers, and pushouts are finite shapes of this one universal law.

boundary A universal object is relative to its diagram and category. It is not an absolute total field containing every context.

source Standard universal-property definition.

2.0.6 H.0025 Pullback and pushout gluing

Theorem · proved-standard

depends: H.0024

statement The pullback $X \times_Z Y$ is terminal among pairs mapping compatibly to X and Y . The pushout $X +_Z Y$ is initial among objects receiving maps from X and Y which agree on Z .

transformations Pullback forms a shared compatible interior. Pushout glues two presented regions along one declared common boundary.

boundary Neither construction licenses gluing from likeness alone; the comparison maps to or from Z are required data.

source Standard finite-limit and finite-colimit universal properties.

2.0.7 H.0026 Span and cospan composition

Theorem · proved-standard

depends: H.0025

statement In a category with pullbacks, spans compose by pullback. In a category with pushouts, cospans compose by pushout. Both compositions are associative up to canonical isomorphism and admit identity spans or cospans.

transformations Spans compare through a common refinement; cospans compose open systems by joining interfaces. They are nonidentical relation species.

boundary Strict associativity requires a chosen strictification or bicategorical treatment; canonical isomorphism is the standard statement.

source Standard span and cospan bicategories.

2.0.8 H.0027 Yoneda relation atlas

Theorem · proved-standard

depends: H.0023

statement For $F : \mathcal{C}^{\text{op}} \rightarrow \text{Set}$,

$$\text{Nat}(\text{Hom}(-, X), F) \equiv F(X)$$

naturally in F and X . Consequently the complete representable functor determines X up to isomorphism.

transformations An object can be recovered from the natural family of all admitted relations to it. This is the precise categorical boundary behind receiver-relative identity.

boundary One hash, scalar, image, cross-ratio, or finite receiver family is not the complete representable functor.

source Yoneda lemma.

2.0.9 H.0028 Adjunction, unit, and counit

Definition · definition

depends: H.0023

statement An adjunction $F \dashv G$ is a natural bijection

$$\text{Hom}_{\mathcal{D}}(FX, Y) \equiv \text{Hom}_{\mathcal{C}}(X, GY).$$

Equivalently it has a unit $\eta : 1 \rightarrow GF$ and counit $\varepsilon : FG \rightarrow 1$ satisfying the two triangle identities.

transformations The unit and counit are opposed passages between two description worlds; their triangles express exact round-trip coherence.

boundary An adjoint is not generally an inverse. The unit or counit may lose distinctions unless it is an isomorphism.

source Standard adjoint-functor definition.

2.0.10 H.0029 Monoidal co-presence

Definition · definition

depends: H.0022

statement A monoidal category carries a bifunctor $\otimes$, a unit object, and coherent associator and unitor isomorphisms. A symmetric monoidal category additionally has a coherent swap $X \otimes Y \equiv Y \otimes X$.

transformations $\otimes$ models co-present composition; categorical composition models succession. Interchanging them requires the monoidal interchange law.

boundary A tensor symbol alone does not specify interaction, entanglement, or physical simultaneity.

source Mac Lane's monoidal-category axioms.

2.0.11 H.0030 Presheaf, sheaf, and exact local gluing

Definition · definition

depends: H.0023, H.0025

statement A presheaf assigns data $F(U)$ contravariantly to regions and restriction maps to inclusions. A sheaf requires that every compatible family on a cover glues to one unique section on the covered region.

transformations Restriction gives receiver-relative local faces; the sheaf axiom states the exact condition under which those faces determine a common interior.

boundary Local agreement without the overlap equations does not imply a global section. Some mathematical data form only presheaves.

source Standard sheaf axiom.

2.0.12 H.0031 Theory, model, and translation portal

Definition · definition

depends: H.0020, H.0023

statement A presented mathematical theory consists of a typed signature, axioms, and a class of models with structure-preserving maps. A translation supplies a map of signatures and an interpretation carrying every source axiom to a theorem of the target.

transformations Every domain enters this synopsis through its presented objects, relations, operations, models, and receivers. Functorial translations can then compare domains without declaring their objects identical.

boundary This portal makes every suitably presented field importable; it is not a proof that holonics is conservative, complete, or equivalent to every mathematical foundation.

source Standard model-theoretic presentation; laboratory atlas schema.

2.0.13 H.0032 Enriched category and structured parameters

Definition · definition

depends: H.0029

statement For a monoidal category $\mathcal{V}$, a $\mathcal{V}$ -enriched category replaces each hom-set by a hom-object $\underline{\text{Hom}}(X, Y) \in \mathcal{V}$ and gives composition and identity as morphisms of $\mathcal{V}$. A parameterized family is a generalized element

$$P \rightarrow \underline{\text{Hom}}(X, Y).$$

transformations The monoidal structure of $\mathcal{V}$ determines how parameters compose. Copying or discarding P requires corresponding comonoid structure; it is not supplied by the word “parameter.”

boundary Enrichment does not choose the parameter category or make every resource duplicable.

source Standard enriched category theory and parameterized circuit semantics.

2.0.14 H.0033 Indexed category, fibration, and Grothendieck assembly

Definition · definition

depends: H.0023

statement An indexed category is a pseudofunctor $F : \mathcal{R}^{\text{op}} \rightarrow \text{Cat}$. Its Grothendieck construction $\int F \rightarrow \mathcal{R}$ is a fibration whose fiber at ρ is $F(\rho)$. Cartesian or cocartesian lifts state how objects transport along arrows of the base.

transformations Receiver-local ecologies live in fibers. The total category assembles their typed transports without becoming a receiver outside the system. Lax or oplax transport retains a directed comparison when strict naturality does not hold.

boundary The Grothendieck total category is not an absolute global state, and a base arrow does not license a lift unless the fibration supplies it.

source Standard indexed-category and Grothendieck-fibration equivalence.

2.0.15 H.0034 Double category and interaction doctrine

Definition · definition

depends: H.0026, H.0029

statement A double category has objects, horizontal arrows, vertical arrows, and squares with horizontal and vertical composition satisfying interchange. A doctrine of open systems may additionally act through an operad or module of interface patterns, separating the systems from the ways their ports, variables, or boundaries interact.

transformations Sequential evolution, boundary rechart, independent juxtaposition, and actual interaction become separately typed compositional directions.

boundary A monoidal coproduct gives juxtaposition only. Contact or co-presence requires a declared interaction pattern and its action.

source Standard double categories, structured cospans, and categorical systems theory.

2.0.16 H.0035 Local sections, global gluing, and obstruction

Theorem · proved-standard

depends: H.0030, H.0033

statement On a site, every section over a covered region restricts to compatible local sections. For a sheaf the converse holds uniquely. Compatible local data with nontrivial overlap holonomy cannot be restrictions of one global section.

transformations “Global over U ” means transported local agreement across a cover of the boundary-defined region U . Nontrivial return around an overlap cycle is a local-to-global obstruction.

boundary The theorem supplies no greatest region and no total field. Pairwise compatibility alone may be insufficient when higher overlaps matter.

source The sheaf axiom and standard Čech obstruction theory.

2.0.17 H.0036 Information order, algebra, and coalgebra

Definition · definition

depends: H.0021, H.0023

statement An information order $x \subseteq y$ says that y extends the distinctions carried by x . An F -algebra $FA \rightarrow A$ folds a recursively generated structure; an F -coalgebra $X \rightarrow FX$ unfolds observable process structure. Well-founded coalgebras support recursion and termination arguments.

transformations Inductive data, streaming state, recursive algorithms, and growing standing can be compared without reducing the process to one endpoint function.

boundary A path witnesses information growth only under the declared domain or synthetic-domain axioms. Not every geometric path is automatically an information extension.

source Standard domain theory, initial-algebra, and coalgebraic recursion.

3 Algorithms, execution, recursion, and rebase

3.0.1 H.0100 Situated parameterized algorithm

Definition · definition

depends: H.0010, H.0011, H.0012, H.0018

statement A situated algorithm is

$$\mathcal{A} = (\Theta, I, S, O, \iota, \delta, \omega, \rho),$$

where Θ is a parameter type; $I_\theta, S_\theta, O_\theta$ are input, state, and output fibers;

$$\iota_\theta \subseteq I_\theta \times S_\theta, \quad \delta_\theta \subseteq S_\theta \times S_\theta,$$

admit initialization and one transition; $\omega_\theta \subseteq S_\theta$ marks completion; and

$$\rho_\theta \subseteq S_\theta \times O_\theta$$

is the output receiver.

receiver The ordinary input/output relation is derived from executions and ρ_θ ; it is not the complete identity of $\mathcal{A}$.

boundary Determinism, totality, termination, computability, reversibility, and complexity are separate predicates, not parts of the word “algorithm.”

source Laboratory synthesis of transition systems, open systems, and receiver semantics.

3.0.2 H.0101 Execution path and trace category

Definition · definition

depends: H.0100

statement At θ , an execution path is a finite word

$$s_0 \rightarrow s_1 \rightarrow \cdots \rightarrow s_n$$

with $\delta_{\theta(s_k, s_{k+1})}$ for every $k < n$. Length-zero paths are identities; concatenation of paths is associative. These paths form the free category generated by the transition quiver δ_θ .

derivation Concatenation is list concatenation with the common endpoint identified. List associativity proves categorical associativity; the singleton endpoint path gives each identity.

transformations A path retains ordered interior occurrences. Relational composition retains only existence of some interior witness.

boundary A trace is not automatically a physical chronology. Its order is the algorithm’s declared transition order.

source Standard free-category construction.

3.0.3 H.0102 Extensional semantics

Definition · definition

depends: H.0100, H.0101

statement The extensional semantics of $\mathcal{A}$ at θ is

$$[\mathcal{A}]_{\theta(i, o)} \leftrightarrow \exists s_0, s_n : \iota_{\theta(i, s_0)} \wedge s_0 \delta_\theta^* s_n \wedge \omega_{\theta(s_n)} \wedge \rho_{\theta(s_n, o)},$$

where δ_θ^* is reflexive-transitive closure.

boundary Equal extensional semantics does not imply equal executions, resource costs, reversible interiors, or causal attribution.

source Standard operational-to-relational semantics.

3.0.4 H.0103 Functional face of a deterministic terminating algorithm

Theorem · proved-derived

depends: H.0100, H.0102

statement If initialization, transition, and observation are single-valued; every initialized execution terminates; and completed states do not transition, then $[\mathcal{A}]_\theta$ is the graph of a partial function $I_\theta \rightarrow O_\theta$. It is total when every input initializes and completes.

derivation Single-valued initialization fixes s_0 . Inductively, single-valued δ_θ fixes every reachable successor. Termination supplies a completed s_n ; terminality prevents a competing later completion; and single-valued ρ_θ fixes o .

boundary The theorem derives a function face; it does not identify the algorithm with that function.

source Elementary transition-system derivation.

3.0.5 H.0104 Lambda

Definition · definition

depends: H.0100, H.0101, H.0102

statement Choose typed ingress and egress boundary receivers

$$\beta_\theta^- : I_\theta \times S_\theta \rightarrow B_\theta^-, \quad \beta_\theta^+ : \text{Path}(\mathcal{A}_\theta) \rightarrow B_\theta^+.$$

The boundary correspondence

$$\Lambda_{\mathcal{A}, \theta}^{\beta^-, \beta^+} \subseteq B_\theta^- \times B_\theta^+$$

relates b^- to b^+ exactly when an admitted execution begins at an initialized occurrence presented by b^- and returns a path face presented by b^+ . One-step, completed, and open-system Lambdas are explicit choices of path class and boundary receivers, not implicit changes in the symbol.

transformations Changing θ may change state fibers, admitted transitions, completion, or receiver. Such change is lawful only with explicit transport between the affected fibers. The historical expression $\Lambda_{F(Q \otimes A \otimes L)} \mapsto (P, \pi_B)$ is one typed specialization when those boundary factors and their maps are supplied; it does not fix the general definition.

boundary The symbol alone asserts no determinism, universality, optimization, or physical law. Omitting $\mathcal{A}, \theta, \beta^-, \beta^+$ leaves an under-specified relation.

source Derived from H.0100–H.0102; laboratory Lambda correction.

3.0.6 H.0105 Sequential composition

Theorem · proved-derived

depends: H.0100, H.0101, H.0102

statement Let $\mathcal{A} : I \rightarrow O$ and $\mathcal{B} : O \rightarrow P$ be situated algorithms whose exposed output/input boundary types agree. Gluing a completed $\mathcal{A}$ path to an initialized $\mathcal{B}$ path gives

$$[\mathcal{B} \circ \mathcal{A}] = [\mathcal{B}] \circ [\mathcal{A}].$$

derivation Expand both sides. Each side is exactly the assertion that there exists an intermediate o and an $\mathcal{A}$ path ending at o followed by an $\mathcal{B}$ path beginning at o . Associativity is reassociation of the witnesses and path concatenation.

transformations Sequential gluing shares one typed boundary. Monoidal co-presence instead forms a product/tensor state and does not impose temporal succession.

boundary A merely equal serialized output is not a typed gluing boundary unless its identification map is supplied.

source Relational composition and free-path concatenation.

3.0.7 H.0106 Invertible rebase is execution conjugacy

Theorem · proved-derived

depends: H.0100, H.0101, H.0102

statement Let $\varphi_\theta : S_\theta \rightarrow S'_\theta$ be a bijection and define

$$\delta'_\theta = \varphi_\theta \circ \delta_\theta \circ \varphi_\theta^{-1}.$$

Transport initialization, completion, and receiver along φ_θ . Then

$$(s_0 \rightarrow \cdots \rightarrow s_n) \mapsto (\varphi s_0 \rightarrow \cdots \rightarrow \varphi s_n)$$

is a length-preserving bijection of executions, and the extensional semantics is unchanged after the corresponding boundary rechart.

derivation For each edge,

$$\delta_{\theta(s_k, s_{k+1})} \leftrightarrow \delta'_{\theta(\varphi s_k, \varphi s_{k+1})}$$

by the definition of conjugacy. Apply this edgewise; φ^{-1} gives the inverse path map. The transported boundary relations make the endpoint semantics commute.

transformations State rebase is conjugacy. Integral change of variables additionally transports measure by a Jacobian and is treated separately.

boundary A noninvertible map is not a rebase. It is a quotient, abstraction, or simulation and may erase paths.

source Elementary conjugacy theorem.

3.0.8 H.0107 Receiver-exact algorithm quotient

Theorem · proved-derived

depends: H.0016, H.0100, H.0102

statement Let $q : S \rightarrow Q$ be surjective. Suppose: (i) initialization factors through q ; (ii) whenever $q(s) = q(t)$, every step from s and t has the same set of q -successor classes; (iii) completion is constant on q -fibers; and (iv) every future receiver factors through q . Then the quotient transition system has exactly the same extensional semantics for those receivers.

derivation Map each source execution through q to obtain a quotient execution. Conversely, lift quotient edges successively using (ii), beginning with an initialized representative from (i). Conditions (iii) and (iv) preserve completion and observation. Thus both semantic relations include each other.

transformations This is the algorithmic form of receiver-exact compression. Bisimulation is a common sufficient carrier for condition (ii).

boundary If only forward simulation is known, one obtains semantic inclusion, not equality. New receivers may invalidate the quotient.

source Standard quotient/bisimulation argument with holonic receiver boundary.

3.0.9 H.0108 Recursion, iteration, and corecursion

Definition · definition

depends: H.0100, H.0101

statement Well-founded recursion defines $f(x)$ from values $f(y)$ on a relation $y \prec x$ admitting no infinite descending chain. Iteration is the orbit

$$x_{n+1} = T(x_n).$$

Corecursion defines an observable stream by repeatedly exposing one output and one successor state.

transformations Each construction is an algorithmic path species. A recursive call graph, an orbit, and a generated stream can share a receiver value without being the same transition system.

boundary A self-referential equation is not a recursion theorem until existence, uniqueness, or productivity is established.

source Standard recursion and coalgebraic definitions.

3.0.10 H.0109 Series as execution lineage and receiver

Definition · definition

depends: H.0108, H.0017

statement Given terms a_n in an additive topological object, define partial states

$$S_0 = 0, \quad S_{n+1} = S_n + a_n.$$

The finite series is the execution lineage (S_n) . An infinite sum $\sum_{n=0}^{\infty} a_n = L$ means $S_n \rightarrow L$ in a declared topology and receiver.

transformations A generating function

$$G(z) = \sum_{n \geq 0} a_n z^n$$

is a receiver of the coefficient lineage. It does not erase the recurrence unless its domain and inversion law are retained.

boundary The glyph L is not the series interior. Rearrangement, analytic continuation, and summability methods require their own hypotheses.

source Standard series definition with laboratory lineage interpretation.

3.0.11 H.0110 Effective computational algorithm

Definition · definition

depends: H.0100, H.0103

statement A computational algorithm is a situated algorithm with finitely or effectively presented state encodings and transition/receiver relations, together with a proved refinement from an executable implementation to the abstract semantics.

receiver Source code, bytecode, circuits, and machine traces are distinct presentations connected by compiler, assembler, and hardware refinement maps.

boundary Church–Turing is a thesis identifying effectively calculable numerical functions with a formal class such as Turing-computable functions. It is not derived merely by defining $\mathcal{A}$.

source Standard semantics/refinement boundary.

3.0.12 H.0111 Relative universal evaluator

Definition · definition

depends: H.0110

statement For an encoding $\text{enc} : \mathcal{C} \rightarrow E$ of a class of algorithms and inputs, a universal evaluator U satisfies

$$U(\text{enc}(A, x)) = [A](x)$$

wherever A terminates. Universality is relative to $\text{cal } C$, its encoding, and the evaluator's operational semantics.

boundary No finite evaluator is an absolute carrier of every mathematical transformation without a declared coding relation.

source Standard universal-machine definition.

4 Algebra, symmetry, order, and combinatorics

4.0.1 H.0120 Algebraic operations and homomorphisms

Definition · definition

depends: H.0020

statement A magma, monoid, group, ring, and field are typed operation systems with successively stronger laws. A homomorphism preserves every operation and constant named by the signature.

transformations Algebraic structure records which compositions are lawful and which identities survive transport.

boundary The same carrier set with different operations is a different algebraic object. “Prime,” “unit,” and “irreducible” depend on that object.

source Standard universal algebra.

4.0.2 H.0121 Group action and orbit-stabilizer

Theorem · proved-standard

depends: H.0120

statement For an action $G \curvearrowright X$, the orbit is $Gx = \{gx : g \in G\}$ and the stabilizer is $G_x = \{g : gx = x\}$. If G is finite, then

$$|Gx| = [G : G_x].$$

transformations An invariant occurrence is fixed by a subgroup; its displayed orbit is the family of faces reached by admitted symmetries.

boundary Equality of orbit invariants need not classify orbits unless a complete invariant theorem is supplied.

source Orbit-stabilizer theorem.

4.0.3 H.0122 Modules, linear maps, and rank-nullity

Theorem · proved-standard

depends: H.0120

statement An R -module carries addition and a compatible R -action. For a linear map $T : V \rightarrow W$ with V finite dimensional,

$$\dim V = \dim \ker T + \dim \text{im } T.$$

transformations The kernel records directions invisible to the receiver T ; the image records the faces it can emit.

boundary Dimension requires the stated scalar and finiteness hypotheses. Rank is not an absolute measure across unrelated modules.

source Standard module theory and rank-nullity theorem.

4.0.4 H.0123 First isomorphism theorem

Theorem · proved-standard

depends: H.0016, H.0122

statement For a module homomorphism $T : M \rightarrow N$, the map

$$\frac{M}{\ker T} \rightarrow \text{im } T, \quad m + \ker T \mapsto T(m)$$

is an isomorphism.

transformations The emitted linear face is exactly the source modulo the distinctions killed by that receiver.

boundary The quotient reconstructs the image, not the original marked occurrences inside each kernel coset.

source First isomorphism theorem for modules.

4.0.5 H.0124 Short exact sequence and splitting

Theorem · proved-standard

depends: H.0123

statement A sequence

$$0 \rightarrow A \xrightarrow{i} B \xrightarrow{p} C \rightarrow 0$$

is exact when i identifies A with $\ker p$ and p is onto. It splits exactly when p has a section, equivalently i has a retraction; then $B \equiv A \oplus C$.

transformations Exactness separates retained interior, total carrier, and exposed quotient. Splitting is the additional recoverability law.

boundary Exactness alone does not supply a canonical decomposition.

source Splitting lemma.

4.0.6 H.0125 Tensor-product universal property

Theorem · proved-standard

depends: H.0029, H.0122

statement For modules M, N , every bilinear map $b : M \times N \rightarrow P$ factors uniquely through a linear map

$$\bar{b} : M \otimes_R N \rightarrow P.$$

transformations The tensor product is the universal linear receiver of bilinear co-presence.

boundary Not every interaction is bilinear, and a tensor product is not sequential composition.

source Universal property of the tensor product.

4.0.7 H.0126 Adjoint, Gram form, and positive factorization

Theorem · proved-standard

depends: H.0122

statement A linear map between finite-dimensional inner-product spaces has a unique adjoint T^* satisfying

$$\langle Tv, w \rangle = \langle v, T^*w \rangle.$$

A finite Hermitian form q is positive semidefinite exactly when $q(v, w) = \langle Sv, Sw \rangle$ for some S .

transformations A positive face is a squared transported distinction. Quotienting by $\ker q$ produces its positive-definite receiver space.

boundary An adjoint is not generally an inverse or a reversal of causation. Positivity must be proved for the complete declared form.

source Finite-dimensional adjoint theorem and Gram factorization.

4.0.8 H.0127 Schur complement and inertia

Theorem · proved-standard

depends: H.0126

statement If $A > 0$, then

$$\begin{pmatrix} A & C \\ C^* & D \end{pmatrix} \geq 0 \leftrightarrow D - C^*A^{-1}C \geq 0.$$

Under congruence, the block matrix has the inertia of A plus the inertia of its Schur complement.

transformations Eliminating one positive block returns an exact effective boundary law and retains the eliminated block's inertia contribution.

boundary Singularity or indefiniteness of A requires a different shorting or generalized-inverse theorem.

source Schur-complement criterion and Sylvester inertia law.

4.0.9 H.0128 Polynomial evaluation and quotient

Theorem · proved-standard

depends: H.0120, H.0123

statement For a commutative ring R and $a \in R$,

$$\ker(\text{ev}_a : R[x] \rightarrow R) = (x - a), \quad \frac{R[x]}{x - a} \cong R.$$

transformations A polynomial presentation and its evaluated value are related through an exact quotient; they are not the same construction.

boundary Evaluation at one point discards every polynomial difference divisible by $x - a$.

source Polynomial remainder theorem and first isomorphism theorem.

4.0.10 H.0129 Chinese remainder decomposition

Theorem · proved-standard

depends: H.0128

statement For pairwise comaximal ideals $I_1, \dots, I_n$,

$$\frac{R}{\{x \in R : \forall i, x \in I_i\}} \cong \prod_i \frac{R}{I_i}.$$

transformations One source admits a compatible product of residue-loop receivers. The inverse is exact only because comaximality supplies the gluing idempotents.

boundary A single residue coordinate does not determine the source class.

source Chinese remainder theorem.

4.0.11 H.0130 Field extension, minimal polynomial, and tower law

Theorem · proved-standard

depends: H.0128

statement An algebraic element α over F has a unique monic minimal polynomial m_α and $F(\alpha) \cong \frac{F[x]}{m_\alpha}$. For finite $F \subset K \subset L$,

$$[L : F] = [L : K][K : F].$$

transformations Adjoining an algebraic axis changes the ambient field by an exact quotient; tower degree records compositional rank.

boundary Transcendental extensions are not finite quotients by a minimal polynomial.

source Standard field-extension theory.

4.0.12 H.0131 Galois correspondence

Theorem · proved-standard

depends: H.0121, H.0130

statement For finite Galois $\frac{L}{F}$ with group G , subgroups $H \leq G$ correspond order-reversingly to intermediate fields through $H \mapsto L^H$ and $K \mapsto \text{Gal}(\frac{L}{K})$. Normal subgroups correspond to Galois subextensions.

transformations Symmetry subgroups and fixed receivers are dual order faces of one field extension.

boundary The finite Galois hypotheses are essential; inseparable and infinite extensions require enlarged forms of the theorem.

source Fundamental theorem of Galois theory.

4.0.13 H.0132 Representations and Schur channels

Theorem · proved-standard

depends: H.0121, H.0122

statement A representation is a homomorphism $G \rightarrow \text{GL}(V)$. An intertwiner commutes with the action. Over an algebraically closed field, finite-dimensional irreducibles V, W obey

$$\text{Hom}_{G(V, W)} = 0$$

when nonisomorphic, and $\text{End}_{G(V)}$ consists of scalars.

transformations Intertwiners are the lawful transports between symmetry-conditioned channels.

boundary Reducible representations and non-algebraically-closed scalars admit larger commutants.

source Schur's lemma.

4.0.14 H.0133 Maschke decomposition

Theorem · proved-standard

depends: H.0132, H.0124

statement If G is finite and $\text{char}(k)$ does not divide $|G|$, every finite-dimensional $k[G]$ -representation is a direct sum of irreducibles.

transformations Averaging a projection over G produces an invariant complement, hence an exact constituent decomposition.

boundary In modular characteristic the averaging denominator vanishes and complete reducibility can fail.

source Maschke's theorem.

4.0.15 H.0140 Poset, meet, join, and lattice

Definition · definition

depends: H.0010

statement A partial order is reflexive, antisymmetric, and transitive. A meet is a greatest lower bound; a join is a least upper bound. A lattice has both for every pair.

transformations Founding, refinement, divisibility, information, and closure orders are different posets until an order-preserving translation relates them.

boundary One chronological order need not totalize co-present or incomparable events.

source Standard order theory.

4.0.16 H.0141 Galois connection and closure

Theorem · proved-standard

depends: H.0140

statement Monotone maps $\alpha : P \rightarrow Q$ and $\gamma : Q \rightarrow P$ form a Galois connection when

$$\alpha(p) \leq q \leftrightarrow p \leq \gamma(q).$$

Then $\gamma\alpha$ is a closure operator and $\alpha\gamma$ an interior operator.

transformations Mutually constrained source and receiver maps induce exact closed and open fixed-point strata.

boundary A feedback pair is not a Galois connection unless the adjunction inequality is proved.

source Standard Galois-connection theorem.

4.0.17 H.0142 Incidence algebra and Möbius inversion

Theorem · proved-standard

depends: H.0140, H.0120

statement On a locally finite poset, convolution is

$$(f \star g)(x, z) = \sum_{x \leq y \leq z} f(x, y)g(y, z).$$

The zeta function has inverse μ . Hence

$$G(x) = \sum_{y \leq x} F(y) \rightarrow F(x) = \sum_{y \leq x} \mu(y, x)G(y).$$

transformations Inclusion over an incidence order and its exact foil are mutually inverse transports.

boundary Local finiteness, or an alternative summability law, is required.

source Möbius inversion on locally finite posets.

4.0.18 H.0143 Formal generating-function convolution

Identity · proved-standard

depends: H.0109, H.0120

statement In the formal power-series ring,

$$[z^n] \left(\sum_{j \geq 0} a_j z^j \right) \left(\sum_{k \geq 0} b_k z^k \right) = \sum_{j=0}^n a_j b_{n-j}.$$

transformations Multiplication composes two counted path populations by every split of total degree n .

boundary Formal validity uses finitely many terms per coefficient and requires no analytic convergence. Evaluation at a number is a later receiver.

source Cauchy product in a formal power-series ring.

4.0.19 H.0144 Graph incidence, cycles, and cuts

Theorem · proved-standard

depends: H.0122, H.0126

statement For an oriented finite graph with incidence matrix B , flows satisfying junction balance form $\ker B$ and potential-generated cuts form $\operatorname{im} B^*$. Over $\mathbb{R}$,

$$(\ker B)^\perp = \operatorname{im} B^*, \quad \operatorname{rank} B = |V| - c$$

where c is the number of connected components.

transformations Cycle current and gradient potential are orthogonal receiver species. Orientation changes signs but not the cycle and cut dimensions.

boundary Weighted, directed-capacity, and nonlinear networks require their own operators; the incidence law alone does not supply dynamics.

source Algebraic graph theory.

4.0.20 H.0145 Maximum flow, minimum cut, and Hall boundary

Theorem · proved-standard

depends: H.0140, H.0144

statement In a finite nonnegative-capacity network, maximum feasible source–sink flow equals minimum cut capacity. A finite bipartite graph has a matching saturating L exactly when

$$|N(S)| \geq |S|$$

for every $S \subseteq L$.

transformations Both theorems replace a global navigation question by an exact family of boundary-capacity conditions.

boundary They do not choose semantic objectives or manufacture absent edges.

source Max-flow/min-cut theorem and Hall’s marriage theorem.

4.0.21 H.0146 Matroid dependence and basis exchange

Definition · definition

depends: H.0140

statement A nonempty family of bases is a matroid basis family when, for $x \in B_1 - B_2$, some $y \in B_2 - B_1$ makes $(B_1 - \{x\}) \cup \{y\}$ a basis. Its rank function is monotone and submodular.

transformations Matroids retain abstract dependence while permitting multiple coordinate bases and exchange paths between them.

boundary Not every context-dependent relation is matroidal; exchange must hold.

source Standard matroid cryptomorphisms.

4.0.22 H.0147 Inclusion–exclusion and binomial transport

Identity · proved-standard

depends: H.0142

statement For finite sets $A_1, \dots, A_n$,

$$|\cup_i A_i| = \sum_{\emptyset \neq J \subseteq \{1, \dots, n\}} (-1)^{|J|+1} |\{x : \forall j \in J, x \in A_j\}|.$$

This is Möbius inversion on the Boolean lattice.

transformations Alternating overlap terms restore distinctions lost by a naive sum of receiver faces.

boundary Infinite families require measure or convergence hypotheses.

source Inclusion–exclusion principle.

4.0.23 H.0148 Linear recurrence and transfer matrix

Theorem · proved-standard

depends: H.0122, H.0143

statement A recurrence

$$a_{n+r} = c_{r-1}a_{n+r-1} + \dots + c_0a_n$$

is equivalent to iteration of its companion matrix on the state $(a_n, \dots, a_{n+r-1})$. Its generating function is rational.

transformations The scalar sequence, finite state transport, and rational series are three exact receiver faces of one recurrence.

boundary Nonlinear, variable-coefficient, and infinite-memory recurrences need different state carriers.

source Standard linear-recurrence theory.

5 Geometry, calculus, series, and invariance

5.0.1 H.0200 Axis, unit, coordinate, and ratio

Definition · definition

depends: H.0002, H.0003

statement An axis is a typed oriented one-dimensional fiber L . A unit is a nonzero local frame $u \in L$. The coordinate of $v \in L$ relative to u is the scalar $[v]_u$ defined by

$$v = [v]_u u.$$

A ratio compares two commensurable quantities after a transport into one common one-dimensional fiber.

transformations If $u' = au$ with $a \neq 0$, then

$$[v]_{u'} = a^{-1}[v]_u.$$

Thus a receiver may normalize any nonzero v to coordinate 1 by selecting $u' = v$; the transition map retains the discarded scale.

boundary The scalar $\frac{1}{2}$ is not meaningful without the two compared axes, their units, and orientation. Coordinate sign may change under orientation reversal while unsigned length does not.

source Elementary one-dimensional linear algebra with holonic unit interpretation.

5.0.2 H.0201 Projective cross-ratio Swing

Definition · definition

depends: H.0200, H.0013

statement For an ordered affine quadruple (A, B, C, D) over a field, first retain the undivided pair

$$N = (C - A)(D - B), \quad D_0 = (C - B)(D - A), \quad \text{Swing}(A, B, C, D) = [N : D_0] \in \mathbb{P}^1.$$

Projective pairs obey

$$[N : D_0] = [\lambda N : \lambda D_0] \quad (\lambda \neq 0).$$

Only on the chart $D_0 \neq 0$ may a quotient receiver report $\chi = \frac{N}{D_0}$. The ordered points, pair, orientation, and chart remain the complete Swing carrier.

transformations Three ordered distinct marks select a projective chart by $(A, B, C) \mapsto (\infty, 0, 1)$; the fourth then has quotient coordinate χ when that chart is admissible. Reordering the marks generally applies one of the six anharmonic transforms rather than preserving the reported quotient.

boundary chi outside this projective carrier must declare its own arguments and invariance law. Equal quotient faces do not identify source occurrences or paths. Four is the local population of one cross-ratio, not a universal bound on receivers.

source Classical projective geometry; Holobrochos conservation and Solve records supply the undivided ratio and path-retention requirement.

5.0.3 H.0202 Möbius invariance of the Swing

Theorem · proved-standard

depends: H.0201

statement For

$$g(z) = \frac{\alpha z + \beta}{\gamma z + \delta}, \quad \alpha\delta - \beta\gamma \neq 0,$$

and every admissible quadruple,

$$\text{Swing}(gA, gB, gC, gD) = \text{Swing}(A, B, C, D) \quad \text{in } \mathbb{P}^1.$$

derivation Direct subtraction gives

$$g(x) - g(y) = \frac{(\alpha\delta - \beta\gamma)(x - y)}{(\gamma x + \delta)(\gamma y + \delta)}.$$

Substitute this identity into the four differences defining the pair. Its two coordinates acquire one common nonzero multiplier: the determinant appears twice and every local chart denominator appears in both coordinates. Projectivization removes that common multiplier. The quotient equality follows only on an admissible denominator chart.

transformations This is exact receiver covariance under PGL_2 . It is the elementary model of a value retained while its coordinate chart changes.

boundary Equal projective Swing classifies ordered distinct quadruples on the projective line up to projectivity; it does not classify arbitrary causal diagrams.

source Classical projective-line theorem; independently rederived above.

5.0.4 H.0203 Simplicial boundary squares to zero

Theorem · proved-standard

depends: H.0010

statement For an oriented n -simplex $[v_0, \dots, v_n]$, define

$$\partial[v_0, \dots, v_n] = \sum_{i=0}^n (-1)^i [v_0, \dots, \hat{v}_i, \dots, v_n].$$

Then $\partial_{n-1} \partial_n = 0$.

derivation Deleting vertices $i < j$ can be performed in two orders. The order i then j has sign $(-1)^{i+j-1}$; the order j then i has sign $(-1)^{i+j}$. Each codimension-two face therefore occurs exactly twice with opposite signs.

transformations This is the exact algebraic carrier of internal-face cancellation.

boundary The identity requires a chain complex and its orientation convention. It is not a universal law attached to every use of the word boundary.

source Classical simplicial homology; independently rederived above.

5.0.5 H.0204 Discrete fundamental theorem and causal parity

Theorem · proved-standard

depends: H.0203, H.0101

statement For a path $v_0 \rightarrow v_1 \rightarrow \dots \rightarrow v_n$ and a potential f ,

$$\sum_{k=0}^{n-1} (f(v_{k+1}) - f(v_k)) = f(v_n) - f(v_0).$$

Every internal event appears once with positive and once with negative orientation.

derivation Expand the sum. Terms $+f(v_k)$ and $-f(v_k)$ cancel for $0 < k < n$, leaving only the oriented boundary values.

transformations Reversing the path negates the boundary. Concatenating two paths cancels their shared endpoint. This is the elementary time/causal-parity carrier over orders of events.

boundary Cancellation supplies conservation for exact differences. A physical constitutive law, storage term, or dissipative source must still be specified in its own domain.

source Finite telescoping identity.

5.0.6 H.0205 Kirchhoff cycle identity for exact potentials

Theorem · proved-standard

depends: H.0204

statement On a closed oriented path $v_n = v_0$,

$$\sum_{k=0}^{n-1} (f(v_{k+1}) - f(v_k)) = 0.$$

Equivalently, the integral of an exact discrete one-form around a cycle vanishes.

derivation Apply H.0204 and use $f(v_n) = f(v_0)$.

transformations This is the pure mathematical core of Kirchhoff's voltage law. A nonzero cycle integral witnesses a non-exact one-form, an enclosed flux, a singularity, or an incompletely represented source.

boundary It does not assert instantaneous physical equilibrium. Storage and dissipation can make a subsystem's boundary flux nonzero over a finite interval while the enlarged closed accounting still balances.

source Graph cohomology / telescoping identity.

5.0.7 H.0206 Chain rule as transport composition

Theorem · proved-standard

depends: H.0105

statement If $f : U \rightarrow V$ is differentiable at x and $g : V \rightarrow W$ is differentiable at $f(x)$, then

$$D(g \circ f)_x = Dg_{f(x)} \circ Df_x.$$

In one dimension,

$$\frac{d}{dx} e^{kx} = k e^{kx}.$$

derivation Write

$$f(x+h) = f(x) + Df_x h + r_{f(h)}, \quad \frac{\|r_{f(h)}\|}{\|h\|} \rightarrow 0,$$

and apply the corresponding expansion for g . The linear term is the composite derivative; every remaining term divided by $\|h\|$ tends to zero.

transformations The factor k records the inner reparameterization. Dropping it collapses the causal chart change.

boundary Differentiability and the topological vector-space hypotheses are required. A symbolic substitution is not automatically differentiable.

source Classical Fréchet chain rule; independently sketched above.

5.0.8 H.0207 Change of variables carries the Jacobian

Theorem · proved-standard

depends: H.0206

statement For an orientation-preserving C^1 diffeomorphism $\varphi : U \rightarrow V$ and integrable f ,

$$\int_V f(y) \, dy = \int_U f(\varphi(x)) |\det D\varphi_x| \, dx.$$

derivation Linear maps scale infinitesimal n -volume by $|\det D\varphi_x|$. The derivative is the exact first-order limit of the increment quotient. Apply its determinant scaling to a refining partition and invoke the measure-theoretic change-of-variables theorem in the limit.

transformations Cartesian-to-polar transport in two dimensions has

$$(x, y) = (r \cos \theta, r \sin \theta), \quad |\det D\varphi| = r,$$

hence $dx \, dy = r \, dr \, d\theta$.

boundary Noninjective maps require multiplicity; singular charts require an appropriate decomposition. Rebase is not measure-preserving unless the Jacobian says so.

source Classical change-of-variables theorem.

5.0.9 H.0208 Fundamental theorem of calculus

Theorem · proved-standard

depends: H.0204, H.0206

statement If f is continuous on $[a, b]$ and

$$F(x) = \int_a^x f(t) \, dt,$$

then $F'(x) = f(x)$ in the interior and

$$\int_a^b f(x) \, dx = F(b) - F(a).$$

derivation The difference quotient is

$$\frac{F(x+h) - F(x)}{h} = \frac{1}{h} \int_x^{x+h} f(t) \, dt.$$

Continuity squeezes this local average to $f(x)$. Summing local differences telescopes to the boundary values, the continuum face of H.0204.

transformations Differentiation and integration are inverse only under the stated regularity and boundary conditions.

boundary A “holonic limit” is not a new limit operation until its topology or convergence receiver is specified.

source Classical FTC, reconstructed from local average plus telescoping.

5.0.10 H.0209 Mean-value and squeeze carriers

Theorem · proved-standard

depends: H.0206, H.0208

statement If f is continuous on $[a, b]$ and differentiable on (a, b) , then some $c \in (a, b)$ satisfies

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

If $g(x) \leq f(x) \leq h(x)$ near a and both outer functions tend to L , then $f(x) \rightarrow L$.

derivation Subtract the secant line from f ; the resulting function has equal endpoint values, so Rolle's theorem gives a zero derivative. For squeeze, every neighborhood of L eventually contains both bounds and therefore the middle value.

transformations On a Riemannian manifold the direct MVT equality generally becomes a geodesic integral or norm inequality after parallel transport. Coordinate values from distinct tangent spaces cannot be subtracted without that transport.

boundary Neither theorem produces a global invariant from local bounds without connectedness, regularity, and receiver compatibility.

source Classical MVT and squeeze theorem.

5.0.11 H.0210 Connection, curvature, and holonomy

Definition · definition

depends: H.0018, H.0206

statement A connection ∇ compares neighboring fibers by covariant differentiation or parallel transport. Its curvature is

$$R(X, Y) = [\nabla_X, \nabla_Y] - \nabla_{[X, Y]}.$$

Holonomy is the automorphism obtained by parallel transport around a closed loop.

transformations Curvature is the infinitesimal residual between two ordered transport paths. Under a change of frame it transforms covariantly; scalar invariants such as traces require additional contraction.

boundary One projected receiver path supplies order, not intrinsic curvature. Curvature requires a connection and comparison of alternate transports.

source Standard differential geometry with comparison-cell interpretation.

5.0.12 H.0211 Finite and convergent geometric series

Theorem · proved-standard

depends: H.0109

statement For every r and integer $N \geq 0$,

$$(1 - r) \sum_{n=0}^N r^n = 1 - r^{N+1}.$$

If $|r| < 1$ in a complete normed field, then

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1 - r}.$$

derivation Multiply the finite sum by $1 - r$; adjacent powers cancel. Under $|r| < 1$, the residual r^{N+1} tends to zero, so the finite identity has the stated limit.

transformations For $r = \frac{1}{2}$ the remainder after N terms is exactly $2^{-(N+1)}$. Zeno's sequence converges because the topology admits this vanishing remainder; no step is erased.

boundary The rational face $\frac{1}{1-r}$ does not replace the partial-sum lineage or license convergence when $|r| \geq 1$.

source Elementary algebra and limit.

5.0.13 H.0212 Exponential series, Euler transport, and logarithmic rebase

Theorem · proved-standard

depends: H.0109, H.0206, H.0211

statement The entire series

$$\exp(z) = \sum_{n=0}^{\infty} \frac{z^n}{n!}$$

satisfies $\exp'(z) = \exp(z)$ and $\exp(z + w) = \exp(z)\exp(w)$. Defining sine and cosine by their power series gives

$$\exp(i\theta) = \cos \theta + i \sin \theta.$$

The positive real value e is $\exp(1)$; π is the least positive θ with $\exp(i\theta) = -1$ under the standard orientation, and 2π is the least positive period.

derivation Termwise differentiation is justified by infinite radius of convergence. The Cauchy product and the binomial theorem give the addition law. Separating even and odd powers of $\exp(i\theta)$ yields the cosine and sine series.

transformations For a logarithm branch,

$$\log_b x = \frac{\ln x}{\ln b}.$$

Changing base is a ratio of two axis coordinates. Complex logarithms are multivalued by $2\pi i$ until a branch cut is selected.

boundary π and e are invariants with many lawful presentations; no one series is their complete identity. Euler's identity is a junction of the exponential, circular, complex, and additive/multiplicative structures, not proof that their constructions are identical.

source Classical power-series construction of exponential and trigonometry.

5.0.14 H.0213 Receiver-relative configuration ecology

Definition · definition

depends: H.0003, H.0018, H.0201, H.0210

statement Let actual co-presence be an incidence object

$$E \rightarrow B_1 \times \cdots \times B_m$$

and let X_e be the configuration participating at $e \in E$. A joint receiver $q_e : X_e \rightarrow \prod_i Y_{i,e}$ and typed comparison laws $b_{a(e, q_e(x))} \bowtie_a 0$ define

$$\Omega_e = \left\{ x \in X_e : b_{a(e, q_e(x))} \bowtie_a 0 \text{ for every active } a \right\}.$$

Boundary equalities stratify

$$\mathcal{X} = \{(e, x) : e \in E, x \in X_e\}.$$

Their discriminant is where the active boundary differentials fail their expected rank.

A receiver- i partial exists only for $V_i \in T_e E$ with $D\tau_{j(V_i)} = 0$ for $j \neq i$. If an active boundary system $B(e, x) = 0$ is regular on a chosen horizontal fiber, its continuation obeys

$$D_x B \nabla_{V_i} x = -(D_e B)[V_i].$$

transformations On an admissible affine chart the Swing satisfies

$$\frac{d\chi}{\chi} = \frac{dA - dC}{A - C} + \frac{dB - dD}{B - D} - \frac{dA - dD}{A - D} - \frac{dB - dC}{B - C}.$$

This covector vanishes on common projective rechart motion, retaining only relative four-member change.

boundary E is not completed to the full product, so entangled receivers need not admit independent partials. Comparison laws and admissible transports come from the studied ecology; the calculus is agnostic, not lawless.

source Laboratory receiver synthesis in standard incidence, stratification, implicit-function, and projective-differential language.

5.0.15 H.0214 Receiver-stratified fundamental theorem

Theorem · proved-derived

depends: H.0208, H.0213

statement Let a receiver path $\gamma : [0, 1] \rightarrow E$ cross finitely many seams t_k and carry chamber transport $\mathcal{P}$. For a transported section S , define the exact seam jump

$$\Delta_k S = S_{t_k^+} - J_k S_{t_k^-}$$

using the declared seam map J_k . Then

$$S_1 - \mathcal{P}_{0 \rightarrow 1} S_0 = \int_0^1 \mathcal{P}_{t \rightarrow 1} \nabla_{\dot{\gamma}} S_t dt + \sum_k \mathcal{P}_{t_k^+ \rightarrow 1} \Delta_k S.$$

derivation Apply the transported fundamental theorem on each smooth chamber, transport every equality into the final fiber, and sum. Interior one-sided terms telescope except for their declared seam differences.

transformations A wholly discrete path has only the exact jump sum. A seam-free path has only covariant transport. Mixed systems retain both without turning discrete events into false infinitesimal instants.

boundary The theorem does not invent seam maps or convergence for infinitely many crossings. A scalar count is a later augmentation which forgets the crossing carrier.

source Piecewise covariant FTC, independently assembled for the receiver ecology.

5.0.16 H.0215 Joint incidence and ordered curvature are distinct second faces

Theorem · proved-derived

depends: H.0013, H.0210, H.0213, H.0214

statement For a receiver square carried into one comparison fiber, the joint second difference is

$$H_{ij}S = S_{11} - S_{10} - S_{01} + S_{00}.$$

With ordered edge transports its curvature residual is

$$\mathcal{R}_{ij} = U_{10,11}U_{00,10} - U_{01,11}U_{00,01}.$$

In the smooth limit,

$$R(V_i, V_j) = [\nabla_{V_i}, \nabla_{V_j}] - \nabla_{[V_i, V_j]}.$$

derivation The first formula is finite inclusion–exclusion; the second subtracts the two paths around the comparison square. The example $S_{ab} = ab$ with identity transports has $H_{ij}S = 1$ and $\mathcal{R}_{ij} = 0$, so joint interaction survives even when order commutes.

transformations If Δ_p is an arithmetic place difference, the mixed commutator

$$\mathcal{C}_{i,p} = \nabla_{V_i}\Delta_p - \Delta_p\nabla_{V_i}$$

compares receiver motion with arithmetic admission.

boundary Nonzero Hessian is not curvature, and zero curvature is not absence of interaction. Both require an actual admitted square; false independent axes may not be added to an entangled incidence base.

source Finite-difference inclusion–exclusion and standard connection curvature.

6 Topology, analysis, dynamics, and fractal iteration

6.0.1 H.0220 Topological space and continuity

Definition · definition

depends: H.0001, H.0005

statement A topology τ on X contains $\emptyset$, X , arbitrary unions of its members, and finite intersections. A map $f : X \rightarrow Y$ is continuous exactly when $f^{-1}(U)$ is open for every open $U \subseteq Y$.

transformations Neighborhoods declare which local continuations a receiver can distinguish. Continuity preserves neighborhood incidence under pullback.

boundary Topology alone supplies no metric, measure, causal order, or rate.

source Standard general topology.

6.0.2 H.0221 Subspace, product, and quotient topology

Theorem · proved-standard

depends: H.0016, H.0024, H.0220

statement The subspace topology is initial for an inclusion; the product topology is initial for coordinate projections; the quotient topology along a surjection $q : X \rightarrow Q$ is final:

$$U \subseteq Q \text{ is open} \leftrightarrow q^{-1}(U) \text{ is open.}$$

transformations Restriction, co-presence, and factorization carry distinct universal topologies.

boundary Quotients may fail separation properties; receiver-exact algebraic factorization does not automatically preserve Hausdorff geometry.

source Standard initial and final topology constructions.

6.0.3 H.0222 Compactness and connectedness

Theorem · proved-standard
depends: H.0220

statement A space is compact when every open cover has a finite subcover and connected when it has no separation into two nonempty disjoint open sets. Continuous images of compact spaces are compact; continuous images of connected spaces are connected.

transformations Compactness can turn an open family of local obligations into a finite subfamily. Connectedness prevents a continuous receiver from splitting one region into a discrete two-face image.

boundary Neither property creates a metric estimate, induction law, or analytic continuation without additional hypotheses.

source Standard topology.

6.0.4 H.0223 Metric, Cauchy lineage, and completion

Definition · definition
depends: H.0220

statement A metric d is nonnegative, symmetric, separates points, and obeys the triangle inequality. A Cauchy sequence satisfies

$$\forall \varepsilon > 0 \exists N \forall m, n \geq N : d(x_m, x_n) < \varepsilon.$$

The completion adjoins equivalence classes of Cauchy sequences and is unique up to unique isometry over the original space.

transformations Completion retains the complete eventual distance relation, not one terminal decimal face.

boundary A metric is a declared receiver law. Different compatible metrics may have the same topology but different geometric scales.

source Standard metric completion theorem.

6.0.5 H.0224 Paths, homotopy, and fundamental group

Definition · definition
depends: H.0101, H.0220

statement A path is a continuous map $[0, 1] \rightarrow X$. Endpoint-fixed homotopy classes of loops at x_0 , under concatenation, form $\pi_1(X, x_0)$.

transformations A returned lineage may be compared modulo continuous deformation while retaining its selected base occurrence.

boundary Homotopy forgets timing, metric, embedding, and much interior geometry. Changing basepoint identifies groups only up to a chosen transport and conjugacy.

source Standard algebraic topology.

6.0.6 H.0225 Covering lift and monodromy

Theorem · proved-standard
depends: H.0224

statement For a covering $p : E \rightarrow X$, every path γ and chosen $e_0 \in p^{-1}(\gamma(0))$ have a unique lift beginning at e_0 . Loops induce an action of $\pi_1(X, x_0)$ on the fiber $p^{-1}(x_0)$.

transformations One received loop transports a constituent among sheets without identifying those sheets.

boundary Branch points, singular seams, and arbitrary multivalued relations are not ordinary coverings.

source Path-lifting theorem and covering monodromy.

6.0.7 H.0226 Homology, cohomology, and Euler characteristic

Definition · definition
depends: H.0122, H.0203

statement A chain complex has $\partial^2 = 0$ and

$$H_{k(C)} = \ker \frac{\partial_k}{\text{im}} \partial_{k+1}.$$

Dual cochains give $H^{k(C;R)}$. For a finite free chain complex,

$$\chi(C) = \sum_{k(-1)}^k \text{rank } C_k = \sum_{k(-1)}^k \text{rank } H_{k(C)}.$$

transformations Boundaries cancel while non-boundary cycles remain as homology classes; cohomology supplies compatible receivers of cycles.

boundary The projective Swing and Euler characteristic are distinct invariants. Homologous chains need not be the same path, embedding, or lineage.

source Standard homological algebra and Euler–Poincaré identity.

6.0.8 H.0227 Relative homology and exact sequence of a pair

Theorem · proved-standard

depends: H.0014, H.0124, H.0226

statement For $A \subseteq X$,

$$C_*(X, A) = C_* \frac{X}{C_*}(A)$$

and there is a natural long exact sequence

$$\cdots \rightarrow H_{k(A)} \rightarrow H_{k(X)} \rightarrow H_{k(X,A)} \xrightarrow{\partial} H_{k-1}(A) \rightarrow \cdots$$

transformations The pair separates an internally admitted body, its exposed subregion, and the relative class that remains.

boundary Exactness records algebraic incidence; it supplies no physical flux or metric by itself.

source Long exact sequence of a topological pair.

6.0.9 H.0240 Measure, integral, and limit exchange

Theorem · proved-standard

depends: H.0019, H.0220

statement A measure is countably additive on a sigma-algebra. Lebesgue integration extends nonnegative simple-function integration. Monotone convergence and dominated convergence give exact hypotheses under which a pointwise limit may pass through the integral.

transformations Integration is a receiver aggregating occurrences while retaining the declared measure. Limit exchange is a theorem, not a symbolic move.

boundary Without monotonicity, domination, uniform integrability, or another exchange theorem, the limit and integral need not commute.

source Lebesgue integration; monotone and dominated convergence theorems.

6.0.10 H.0241 Normed, Banach, Hilbert, and L_p spaces

Definition · definition

depends: H.0223, H.0240

statement A Banach space is a complete normed vector space. A Hilbert space is complete under a norm induced by an inner product. For $1 \leq p < \infty$, L^p identifies functions equal almost everywhere under

$$\|f\|_p = \left(\int |f|^p \right)^{\frac{1}{p}}.$$

transformations Choosing norm, weak, pointwise, or measure topology selects a different exact convergence receiver.

boundary Almost-everywhere quotienting is exact for measure receivers but erases pointwise distinctions on null sets.

source Standard functional analysis and measure theory.

6.0.11 H.0242 Riesz representation in Hilbert space

Theorem · proved-standard

depends: H.0241

statement Every continuous linear functional ℓ on a Hilbert space H has a unique $y \in H$ with

$$\ell(x) = \langle x, y \rangle, \quad \|\ell\| = \|y\|.$$

transformations A continuous scalar receiver is represented by an internal carrier relative to the Hilbert metric.

boundary This representation is Hilbert-specific; arbitrary Banach duals need not be represented by members of the original space.

source Riesz representation theorem for Hilbert spaces.

6.0.12 H.0243 Self-adjoint spectral theorem and positivity

Theorem · proved-standard

depends: H.0242

statement A bounded self-adjoint operator A has a unique projection-valued measure E such that

$$A = \int_{\sigma(A)} \lambda dE(\lambda).$$

Moreover $A \geq 0$ exactly when $\sigma(A) \subseteq [0, \infty)$, exactly when $\langle Ax, x \rangle \geq 0$ for every x .

transformations Spectrum and quadratic response are exact receiver faces of one self-adjoint operator.

boundary Spectrum alone does not reconstruct a general nonnormal operator. Unbounded operators require explicit dense domains.

source Spectral theorem for bounded self-adjoint operators.

6.0.13 H.0244 Positive kernel and reproducing-kernel space

Theorem · proved-standard

depends: H.0126, H.0242

statement A Hermitian kernel $K : X \times X \rightarrow \mathbb{C}$ is positive definite when

$$\sum_{i,j} \bar{c}_i c_j K(x_i, x_j) \geq 0$$

for every finite population. It determines a unique reproducing-kernel Hilbert space up to canonical isometry.

transformations Pair relations become a Gram geometry in which evaluation is represented internally.

boundary Sampled positivity or finitely many matrix orders does not establish the all-populations condition.

source Moore–Aronszajn theorem.

6.0.14 H.0245 Distributions and weak derivatives

Definition · definition

depends: H.0240

statement A distribution is a continuous linear functional on a test-function space. Its derivative is defined by

$$\langle \partial_i T, \varphi \rangle = -\langle T, \partial_i \varphi \rangle.$$

transformations Differentiation transfers across the pairing, allowing singular or nonsmooth currents to retain exact differential incidence.

boundary A distribution need not be a pointwise function or a positive measure.

source Schwartz distribution theory.

6.0.15 H.0246 Fourier–Plancherel and convolution transport

Theorem · proved-standard

depends: H.0143, H.0207, H.0241

statement With a fixed normalization, the Fourier transform extends uniquely to a unitary map on $L^2(\mathbb{R}^n)$:

$$\|\hat{f}\|_2 = \|f\|_2.$$

If $f, g \in L^1(\mathbb{R}^n)$, $\widehat{f \star g} = \hat{f} \hat{g}$.

transformations Spatial occurrence and frequency occurrence are exact dual receiver faces; convolution becomes pointwise composition.

boundary A sampled discrete transform or floating array is not the complete transform of the source function.

source Plancherel theorem and Fourier convolution theorem.

6.0.16 H.0247 Poisson summation

Theorem · proved-standard

depends: H.0246

statement For a Schwartz function f on $\mathbb{R}$ and the compatible Fourier normalization,

$$\sum_{n \in \mathbb{Z}} f(n) = \sum_{k \in \mathbb{Z}} \hat{f}(k).$$

transformations A lattice population and its reciprocal-lattice spectrum are exact dual sums. Scaling the lattice transports the dual scale reciprocally.

boundary Weaker functions require explicit summability or distributional hypotheses.

source Poisson summation formula.

6.0.17 H.0248 Bochner spectral representation

Theorem · proved-standard

depends: H.0244, H.0246

statement A continuous positive-definite function $\varphi : \mathbb{R}^n \rightarrow \mathbb{C}$ is exactly the Fourier–Stieltjes transform of a finite positive Borel measure:

$$\varphi(x) = \int \exp(ix \cdot \xi) \, d\mu(\xi).$$

transformations A translation-covariant recurrence field has a positive spectral population precisely when its complete kernel is positive definite.

boundary Positivity on selected displacements or finite samples does not establish global positive definiteness.

source Bochner’s theorem.

6.0.18 H.0249 Laplace and Mellin receivers

Definition · definition

depends: H.0207, H.0240

statement Where the integrals converge,

$$\mathcal{L}f(s) = \int_0^\infty \exp(-st)f(t) \, dt, \quad \mathcal{M}f(s) = \int_0^\infty f(x)x^{s-1} \, dx.$$

In particular,

$$\Gamma(s) = \mathcal{M}(\exp(-x))(s) \quad (\operatorname{Re} s > 0).$$

The logarithmic substitution $x = \exp(u)$ turns Mellin transport into a two-sided Laplace/Fourier-type transport with its Jacobian.

transformations Laplace receives additive time; Mellin receives multiplicative scale. Their domains of convergence are part of the face.

boundary Formal symbolic transforms do not establish convergence, inversion, or analytic continuation.

source Classical Laplace and Mellin transform theory.

6.0.19 H.0250 Exact convergence and closure taxonomy

Boundary · proved-derived

depends: H.0005, H.0109, H.0220, H.0241

statement Keep distinct coefficientwise equality of formal series; pointwise, uniform, metric, norm, weak, measure, and operator convergence; topological closure; asymptotic expansion; and certified enclosure. Each is an exact relation with its own quantifiers.

transformations A theorem may transport one convergence species to another only under stated hypotheses, such as domination, compactness, boundedness, or uniformity.

boundary Matching digits, sampled agreement, and visual stability imply none of these relations without a sound certificate.

source Standard analysis taxonomy under the laboratory exactness boundary.

6.0.20 H.0251 Nested exact enclosures

Theorem · proved-standard

depends: H.0222, H.0223, H.0250

statement If rational closed intervals $I_n = [a_n, b_n]$ satisfy $I_{n+1} \subseteq I_n$, every I_n contains x , and $b_n - a_n \rightarrow 0$, then

$$\{y : \forall n, y \in I_n\} = \{x\}.$$

transformations Each finite enclosure is a certified face; nestedness and vanishing width retain the exact law connecting those faces to one real value.

boundary Printing a floating midpoint without the interval and proof discards the certificate.

source Nested-interval theorem.

6.0.21 H.0252 Picard–Lindelöf and local flow

Theorem · proved-standard

depends: H.0206, H.0223

statement If $f(t, x)$ is continuous in t and locally Lipschitz in x , then

$$\dot{x} = f(t, x), \quad x(t_0) = x_0$$

has a unique local solution with continuous dependence on initial data. An autonomous law gives $\varphi_{t+s} = \varphi_t \circ \varphi_s$ where defined.

transformations Law, initial occurrence, and elapsed parameter jointly determine the local lineage.

boundary Non-Lipschitz seams may branch; local existence does not imply global completeness.

source Picard–Lindelöf theorem.

6.0.22 H.0253 Dynamical system, orbit, conjugacy, and factor

Definition · definition

depends: H.0106, H.0220

statement A discrete dynamical system is (X, T) with iterates T^n . A conjugacy h is a homeomorphism satisfying $hT = Sh$. A factor map is a continuous surjection q with $qT = Sq$.

transformations Conjugacy is invertible rebase of complete orbit structure; a factor is a quotient receiver and may discard distinctions.

boundary Matching some orbit statistics does not prove conjugacy or factorhood.

source Standard topological dynamics.

6.0.23 H.0254 Poincaré recurrence

Theorem · proved-standard

depends: H.0019, H.0240, H.0253

statement If T preserves a finite measure μ , then for every measurable A , almost every $x \in A$ returns to A infinitely often.

transformations Recurrence is conditioned by a finite invariant measure; it is not ontological chance.

boundary The theorem supplies no return-time bound, convergence, periodicity, or mixing.

source Poincaré recurrence theorem.

6.0.24 H.0255 Birkhoff ergodic theorem

Theorem · proved-standard

depends: H.0109, H.0254

statement For a measure-preserving T on a probability space and $f \in L^1$,

$$\frac{1}{N} \sum_{k=0}^{N-1} f \circ T^k \rightarrow \mathbb{E}[f \mid \mathcal{I}]$$

almost everywhere, where $\mathcal{I}$ is the invariant sigma-algebra. Under ergodicity the limit is $\int f \, d\mu$.

transformations A long recurrence lineage may have an exact conditional-invariant aggregate face.

boundary The aggregate erases order and implies neither mixing nor finite-time predictability.

source Birkhoff pointwise ergodic theorem.

6.0.25 H.0256 Contraction fixed point and self-similar attractor

Theorem · proved-standard

depends: H.0108, H.0223

statement A contraction on a complete metric space has a unique fixed point and all iterates converge to it. For finitely many contractions f_i , the Hutchinson map

$$\mathcal{F}(K) = \cup_i f_{i(K)}$$

is a contraction on nonempty compact sets in Hausdorff distance and has a unique fixed compact set.

transformations Exact self-similarity consists of named rebase maps, a return law, and convergence in a declared hyperspace metric.

boundary Visual recurrence does not establish a contraction system. Dimension formulas require separation hypotheses.

source Banach fixed-point theorem; Hutchinson self-similarity theorem.

6.0.26 H.0257 Fatou–Julia partition and critical orbit

Theorem · proved-standard

depends: H.0220, H.0253

statement For a rational map R of degree at least 2, the Fatou set is the maximal normality region and the Julia set its completely invariant perfect complement; repelling periodic points are dense in the Julia set. For $z^2 + c$, the Julia set is connected exactly when the critical orbit of 0 is bounded.

transformations Orbit, stability region, critical seam, and parameter world are distinct linked receivers.

boundary A fractal rendering supplies no arithmetic theorem without a proved conjugacy or invariant map.

source Classical Fatou–Julia theory.

6.0.27 H.0258 Partial differential operator and weak solution

Definition · definition

depends: H.0245

statement A linear second-order scalar PDE has principal symbol determined by its highest derivatives; its elliptic, hyperbolic, or parabolic type is classified from that symbol under the corresponding nondegeneracy hypotheses. A weak solution satisfies the equation after transport of derivatives onto test functions.

transformations The principal symbol determines local characteristic geometry; the weak receiver retains boundary and incidence without requiring pointwise differentiability.

boundary Classification does not itself supply existence, uniqueness, regularity, or boundary conditions.

source Standard PDE and distribution theory.

6.0.28 H.0259 Heat semigroup and exact dissipation

Theorem · proved-standard

depends: H.0243, H.0258

statement For a sufficiently regular square-integrable field on Euclidean $\mathbb{R}^n$ with decay sufficient for integration by parts,

$$\partial_t u = \Delta u, \quad u(t) = \exp(t\Delta)u_0,$$

and

$$\frac{d}{dt}\|u(t)\|_2^2 = -2\|\nabla u(t)\|_2^2.$$

transformations Diffusion is geometry-conditioned transport with one exact monotone energy receiver.

boundary A manifold version additionally depends on its metric and boundary conditions. The semigroup generally does not preserve reconstructible high-frequency detail backward in time.

source Spectral heat-semigroup theory and integration by parts.

6.0.29 H.0260 Wave energy and finite propagation

Theorem · proved-standard

depends: H.0246, H.0258

statement For sufficiently regular compactly supported solutions of

$$u_{tt} - c^2 \Delta u = 0,$$

the energy

$$E(t) = \frac{1}{2} \int (|u_t|^2 + c^2 |\nabla u|^2) \, dx$$

is constant, and the value at (t, x) depends only on initial data within $|y - x| \leq c|t|$.

transformations The light cone is the exact domain of dependence of this hyperbolic law; conservation belongs to the complete evolving solution.

boundary Finite propagation is not universal: heat transport has a different support law, and changing the operator changes the cone.

source Classical wave-equation energy identity and domain of dependence.

6.0.30 H.0261 Exact asymptotic relation

Definition · definition

depends: H.0005, H.0250

statement $f(x) = O(g(x))$ as $x \rightarrow a$ means that there exist exact constants $C > 0$ and a neighborhood on which $|f(x)| \leq C|g(x)|$. $f = o(g)$ means

$$\forall \varepsilon > 0 \exists U \forall x \in U : |f(x)| \leq \varepsilon |g(x)|.$$

transformations Asymptotic notation is a quantified bound receiver, not a floating fit.

boundary It does not identify leading coefficients or preserve signs unless those properties are separately proved.

source Standard asymptotic notation.

7 Manifolds, curvature, projection, knots, and simplicial geometry

7.0.1 H.0270 Smooth manifold, tangent space, and differential

Definition · definition

depends: H.0206, H.0220

statement A smooth n -manifold has compatible charts into $\mathbb{R}^n$. A smooth map $f : M \rightarrow N$ induces

$$df_p : T_p M \rightarrow T_{f(p)} N, \quad d(g \circ f)_p = dg_{f(p)} \circ df_p.$$

transformations Charts are local receiver faces; the differential is their exact first-order transport.

boundary A smooth structure supplies no lengths, angles, preferred coordinates, or connection.

source Standard differential geometry.

7.0.2 H.0271 Differential forms and exterior derivative

Identity · proved-standard

depends: H.0203, H.0270

statement A differential k -form is a section of $\Lambda^k T^*M$. The exterior derivative is the unique graded derivation extending the differential of functions and satisfying

$$d^2 = 0, \quad d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^{\deg \alpha} \alpha \wedge d\beta.$$

transformations Forms receive oriented k -dimensional transport; $d^2 = 0$ is the smooth counterpart of boundary cancellation.

boundary Closed need not mean exact. The identity does not manufacture an interior for every closed form.

source Exterior-calculus theorem.

7.0.3 H.0272 Stokes theorem

Theorem · proved-standard

depends: H.0208, H.0271

statement For an oriented compact smooth n -manifold M with boundary and a smooth $(n-1)$ -form ω ,

$$\int_M d\omega = \int_{\partial M} \omega$$

with the induced boundary orientation.

transformations Interior differential and exposed boundary testimony are exact receivers of one oriented body.

boundary A conservation statement follows only after the relevant closedness or source equation is supplied.

source Generalized Stokes theorem.

7.0.4 H.0273 Projective space, incidence, and recharting

Theorem · proved-standard

depends: H.0201, H.0202, H.0270

statement $\mathbb{P}(V) = \frac{V - \{0\}}{\mathbb{R}^\times}$. Standard affine charts are related by rational transition maps; $\mathrm{PGL}(V)$ preserves projective incidence, and on a projective line it preserves the Swing of H.0201.

transformations Homogeneous rescaling changes a coordinate presentation while retaining the projective occurrence and its incidence.

boundary Projective maps need not preserve Euclidean distance, angle, area, or causal path.

source Classical projective geometry.

7.0.5 H.0274 Riemannian metric and Levi-Civita connection

Theorem · proved-standard

depends: H.0210, H.0270

statement A Riemannian metric is a smooth positive-definite symmetric form g_p on every $T_p M$. There is a unique connection ∇ which is metric-compatible and torsion-free:

$$\nabla g = 0, \quad \nabla_X Y - \nabla_Y X = [X, Y].$$

transformations A local unit law and its path transport are separate structures coupled canonically after g is declared.

boundary Positivity is specific to Riemannian geometry. Lorentzian and other indefinite metrics have different cone and sign structure.

source Fundamental theorem of Riemannian geometry.

7.0.6 H.0275 Geodesic and exponential map

Theorem · proved-standard

depends: H.0252, H.0274

statement For $p \in M$ and $v \in T_p M$, there is a unique local geodesic with

$$\nabla_{\dot{\gamma}} \dot{\gamma} = 0, \quad \gamma(0) = p, \quad \dot{\gamma}(0) = v.$$

The exponential map is a diffeomorphism from a neighborhood of $0 \in T_p M$ to a normal neighborhood of p .

transformations Local straight continuation is determined by the connection rather than an external Cartesian frame.

boundary This is local. Cut loci, conjugate points, incompleteness, and multiple global geodesics remain possible.

source Geodesic existence theorem and normal-neighborhood theorem.

7.0.7 H.0276 Curvature, Jacobi deviation, and holonomy

Theorem · proved-standard

depends: H.0210, H.0274, H.0275

statement Curvature is

$$R(X, Y)Z = \nabla_X(\nabla_Y Z) - \nabla_Y(\nabla_X Z) - \nabla_{[X, Y]}Z.$$

A geodesic variation has Jacobi field

$$\nabla_t^2 J + R(J, \dot{\gamma})\dot{\gamma} = 0.$$

Ambrose–Singer generates the holonomy Lie algebra from transported curvature endomorphisms.

transformations Curvature is the residual between neighboring transports; Jacobi fields receive relative continuation; holonomy is a returned finite path face.

boundary One loop or projected crossing does not determine the complete connection or curvature tensor.

source Jacobi equation and Ambrose–Singer theorem.

7.0.8 H.0277 de Rham correspondence

Theorem · proved-standard

depends: H.0226, H.0271

statement Integration of forms over singular chains induces a natural isomorphism

$$H_{\text{dR}}^k(M) \equiv H_{\text{sing}}^k(M; \mathbb{R}).$$

transformations Local differential transport and global incidence topology are exact alternate presentations of one cohomology class.

boundary The theorem identifies classes, not individual forms or path interiors. Harmonic representatives require a metric and Hodge theory.

source de Rham theorem.

7.0.9 H.0278 Gauss–Bonnet curvature total

Theorem · proved-standard

depends: H.0226, H.0272, H.0274, H.0276

statement For a closed oriented Riemannian surface,

$$\int_M K \, dA = 2\pi\chi(M).$$

transformations Distributed local Gaussian curvature integrates to one topological receiver invariant.

boundary This exact formula is two-dimensional. A surface with boundary owes geodesic-curvature and corner terms. It does not identify the projective Swing with Euler characteristic.

source Gauss–Bonnet theorem.

7.0.10 H.0279 Ricci flow and metric response

Theorem · proved-standard
depends: H.0252, H.0274, H.0276

statement On a closed smooth manifold, an initial metric g_0 admits a unique short-time solution

$$\partial_t g = -2 \operatorname{Ric}(g), \quad g(0) = g_0,$$

and its volume form obeys

$$\partial_t(d\mu_g) = -Rd\mu_g.$$

transformations The metric receiver itself evolves by contracted curvature instead of observing a fixed geometry from outside.

boundary Singular continuation, surgery, convergence, and topology classification require additional theorems.

source Hamilton’s short-time Ricci-flow theorem.

7.0.11 H.0280 Three-dimensional Poincaré theorem

Theorem · proved-standard
depends: H.0224, H.0279

statement Every closed connected simply connected topological 3-manifold is homeomorphic to S^3 .

transformations Complete global return structure constrains topology, but the proof passes through controlled geometric evolution and singularity analysis.

boundary This dimension-specific theorem is not a consequence of the phrase “all loops close” or of short-time Ricci flow alone.

source Perelman’s proof through Ricci flow with surgery.

7.0.12 H.0281 Symplectic structure and Darboux chart

Theorem · proved-standard
depends: H.0271

statement A symplectic form is a closed nondegenerate 2-form ω . Locally there are coordinates with

$$\omega = \sum_{j=1}^n dq_j \wedge dp_j.$$

transformations Every local symplectic region has the same incidence normal form while its global winding and topology may differ.

boundary Symplectic structure supplies no metric, positive energy, or preferred global split.

source Darboux theorem.

7.0.13 H.0282 Hamiltonian flow and Liouville preservation

Theorem · proved-standard
depends: H.0281

statement Define X_H by $\iota_{X_H} \omega = dH$. Then

$$\mathcal{L}_{X_H} \omega = 0.$$

An autonomous H is constant along its own flow, and the flow preserves the symplectic volume $\frac{\omega^n}{n!}$.

transformations Hamiltonian current preserves the symplectic relation and its volume across time.

boundary A time-dependent Hamiltonian need not be conserved; symplectic preservation supplies no dissipation law.

source Cartan formula and Liouville theorem.

7.0.14 H.0283 Holomorphic local degree and argument principle

Theorem · proved-standard
depends: H.0224, H.0261, H.0270

statement A nonconstant holomorphic map between Riemann surfaces has local form

$$f(z) - f(p) = a(z - p)^m + O((z - p)^{m+1}), \quad a \neq 0.$$

For meromorphic f with no boundary zero or pole,

$$\frac{1}{2\pi i} \int_{\partial D} \frac{f'}{f} dz = N_{D(f)} - P_{D(f)}$$

with multiplicity.

transformations Local branching has integer degree; boundary winding exactly counts interior zero and pole incidence.

boundary Winding counts but does not locate or metrically classify the individual singularities.

source Local mapping theorem and argument principle.

7.0.15 H.0284 Conformal maps and Möbius automorphisms

Theorem · proved-standard

depends: H.0202, H.0283

statement A holomorphic map with nonzero derivative preserves oriented angles. Automorphisms of the Riemann sphere are Möbius maps, and automorphisms of the unit disk have the form

$$\exp(i\theta) \frac{z - a}{1 - \bar{a}z}, \quad |a| < 1.$$

transformations The receiver may rechart a face conformally while the cross-ratio and complex orientation retain exact projective information.

boundary Conformal maps preserve angles, not general lengths or areas.

source Classical one-complex-variable automorphism theorems.

7.0.16 H.0285 Hyperbolic metric and law of cosines

Theorem · proved-standard

depends: H.0200, H.0274

statement In curvature -1 , a geodesic triangle with side lengths a, b, c and angle γ opposite c obeys

$$\cosh c = \cosh a \cosh b - \sinh a \sinh b \cos \gamma.$$

The disk and upper-half-plane models are related by a Möbius isometry.

transformations A triangle law changes with ambient curvature while projective or conformal recharting can preserve its hyperbolic metric relation.

boundary Euclidean, spherical, and hyperbolic cosine laws are not interchangeable. Gyrovector notation is one model-specific re-expression.

source Hyperbolic law of cosines and Cayley transform.

7.0.17 H.0286 Conics, quadratic forms, and focal receivers

Definition · definition

depends: H.0122, H.0126, H.0273

statement A projective conic is the zero locus of a nonzero homogeneous quadratic form. In an affine Euclidean chart, nondegenerate real conics appear as ellipses, hyperbolas, or parabolas according to the restricted form and chart; circles are a metric subclass of ellipses.

transformations Foci, axes, tangents, and projective incidence are different receiver structures on one quadratic locus. Changing chart may change its affine species while retaining projective nonsingularity.

boundary A focal point or visible arc is not an intrinsic projective invariant.

source Classical projective classification of conics.

7.0.18 H.0287 Embedded knot and projected diagram

Definition · definition

depends: H.0220, H.0270

statement A knot is a tame embedding $K : S^1 \rightarrow M^3$. A generic projection to a surface has finitely many transverse double points supplied with over/under data.

transformations Knot type belongs to the embedding; crossing data belongs to a declared projection receiver.

boundary A projected double point is not a source self-intersection or intrinsic knot vertex.

source Standard knot and regular-diagram definitions.

7.0.19 H.0288 Reidemeister presentation equivalence

Theorem · proved-standard
depends: H.0287

statement Two plane or sphere diagrams represent ambient-isotopic links in $\mathbb{R}^3$ or S^3 exactly when they are related by planar or spherical isotopy and finitely many Reidemeister I, II, and III moves.

transformations The three local moves generate equality of regular projection presentations for one link type.

boundary A crossing change is not a Reidemeister move and may alter knot type.

source Reidemeister theorem.

7.0.20 H.0289 Alexander braid presentation

Theorem · proved-standard
depends: H.0287

statement Every oriented link in S^3 is isotopic to the closure of a braid.

transformations A closed return can be re-presented as ordered transport about a selected axis.

boundary The braid axis and braid word are presentation data, not intrinsic link identity.

source Alexander's braid theorem.

7.0.21 H.0290 Markov closure equivalence

Theorem · proved-standard
depends: H.0289

statement Two braids have isotopic oriented closures exactly when they are connected by braid relations, conjugation, and stabilization or destabilization.

transformations Distinct axis-relative ordered lineages may close to the same standing link.

boundary Markov equivalence need not preserve braid index, word length, crossing count, or construction path.

source Markov theorem.

7.0.22 H.0291 Prime-knot decomposition

Theorem · proved-standard
depends: H.0287

statement Every nontrivial knot in S^3 is a finite connected sum of prime knots, uniquely up to order and isotopy.

transformations Irreducibility is typed by the connected-sum composition law.

boundary Prime-knot factorization does not imply additive crossing number, unknotting number, substitution cost, or arithmetic primality.

source Schubert prime decomposition theorem.

7.0.23 H.0292 Tangle composition and skein quotient

Definition · definition
depends: H.0025, H.0125, H.0288

statement A tangle is an embedded 1-manifold with marked boundary and composes by compatible boundary gluing. A skein module is the free module on isotopy classes modulo a declared family of local linear relations.

transformations Tangled interiors expose typed interfaces; a skein evaluation is receiver-exact when it factors through the declared quotient.

boundary A skein relation is neither ambient-isotopy equality nor positivity without a separate theorem.

source Standard tangle category and skein-module construction.

7.0.24 H.0293 Receiver-visible crossing set

Definition · proved-derived
depends: H.0003, H.0287

statement For an embedding $k : \Gamma \rightarrow M^3$ of a one-complex and a generic projection receiver $\rho : M^3 \rightarrow \Sigma$, define

$$\text{Cross}_{\rho(k)} = \{(x, y) \in \Gamma \times \Gamma : x \neq y, \rho(k(x)) = \rho(k(y))\}$$

together with depth or over/under ordering. This set varies with ρ while ambient isotopy class is separately retained.

transformations Preprocessing the receiver changes visible crossing events at discriminant orientations where genericity fails. In the classical plane/sphere setting, Reidemeister moves classify the resulting local diagram changes.

boundary Crossing count is a receiver statistic, not a complete knot invariant or a proof of arithmetic irreducibility.

source Laboratory ray-receiver definition grounded in regular projection theory.

7.0.25 H.0294 Simplicial complex, realization, and subdivision

Definition · definition

depends: H.0203, H.0220

statement An abstract simplicial complex is a vertex family closed under taking subsets. Its geometric realization glues standard simplices along common faces. Barycentric subdivision produces a canonically homeomorphic realization and induces the same homology.

transformations Triangular and higher simplicial faces can hinge only through declared shared faces. Subdivision changes presentation grain while retaining topology.

boundary A simplicial complex alone supplies no update law, metric, orientation, probability, or physical dynamics.

source Standard simplicial topology.

7.0.26 H.0295 Hausdorff measure and fractal dimension

Definition · definition

depends: H.0223, H.0250

statement The s -dimensional Hausdorff measure is the limit of infimal $\sum_i (\text{diam } U_i)^s$ over covers with diameters bounded by a grain tending to zero. The Hausdorff dimension is the critical s where the measure changes from infinity to zero.

transformations Scaling complexity is received through all covering grains, not through a single rendered zoom.

boundary Hausdorff dimension does not classify fractals, dynamics, or arithmetic structure by itself.

source Hausdorff measure and dimension.

8 Counting, primes, analytic transport, and completion

8.0.1 H.0300 Divisibility, unit, irreducible, and prime

Definition · definition

depends: H.0001

statement In an integral domain R , a divides b when $b = ac$ for some $c \in R$. A unit has a multiplicative inverse. A nonzero nonunit p is irreducible when $p = ab$ implies that a or b is a unit. It is prime when p divides ab implies p divides a or p divides b .

transformations In the integers, primes and irreducibles agree. In a general integral domain they need not; prime always implies irreducible, while the converse requires additional structure such as a unique factorization domain.

boundary Prime is relative to the multiplication and units of the ambient domain. A prime integer, prime ideal, prime knot, and primitive closed orbit are distinct irreducibility species.

source Standard commutative algebra.

8.0.2 H.0301 Euclid's lemma

Theorem · proved-standard

depends: H.0300

statement For a prime integer p , if p divides ab , then p divides a or p divides b .

derivation If p does not divide a , then $\gcd(p, a) = 1$. Bézout gives $up + va = 1$. Multiplying by b yields

$$ubp + vab = b.$$

Both terms on the left are divisible by p , so p divides b .

boundary The Bézout step uses the Euclidean/PID structure of the integers.

source Classical elementary number theory.

8.0.3 H.0302 Fundamental theorem of arithmetic

Theorem · proved-standard

depends: H.0301

statement Every integer $n > 1$ has a factorization

$$n = \prod_p p^{v_p(n)}$$

with finite support, unique up to ordering.

derivation Existence follows by strong induction: if n is not prime, write $n = ab$ with $1 < a, b < n$ and factor both. For uniqueness, if $p_1 \cdots p_r = q_1 \cdots q_s$, Euclid's lemma makes p_1 divide some q_j ; primality forces equality. Cancel and induct.

transformations The map $n \mapsto \left(v_{p(n)} \right)_p$ identifies positive rational numbers with the direct sum of integer valuation axes:

$$v_{p(xy)} = v_{p(x)} + v_{p(y)}.$$

boundary The valuation vector is exact for multiplicative arithmetic. Decimal digits and radix endings are receiver charts, not prime ontology.

source Classical fundamental theorem of arithmetic.

8.0.4 H.0303 Arithmetic convolution and Möbius function

Definition · definition

depends: H.0302

statement For arithmetic functions f, g define Dirichlet convolution

$$(f \star g)(n) = \sum_{d|n} f(d)g\left(\frac{n}{d}\right).$$

Its identity is $\varepsilon(1) = 1, \varepsilon(n) = 0$ for $n > 1$. The Möbius function is

$$\mu(n) = \begin{cases} 0 & \text{if a square prime divides } n \\ (-1)^r & \text{if } n \text{ is a product of } r \text{ distinct primes.} \end{cases}$$

boundary Arithmetic Möbius inversion, incidence-algebra Möbius inversion, and a Möbius fractional-linear transformation are different objects.

source Standard multiplicative number theory.

8.0.5 H.0304 Möbius inversion

Theorem · proved-standard

depends: H.0303

statement Let $u(n) = 1$. Then

$$\mu \star u = \varepsilon.$$

Consequently,

$$F(n) = \sum_{d|n} f(d) \leftrightarrow f(n) = \sum_{d|n} \mu(d)F\left(\frac{n}{d}\right).$$

derivation By multiplicativity it suffices to evaluate a prime power:

$$\sum_{d|p^k} \mu(d) = \mu(1) + \mu(p) = 0$$

for $k \geq 1$, while the sum is 1 at $n = 1$. Convolving $F = f \star u$ with μ gives

$$F \star \mu = f \star (u \star \mu) = f.$$

transformations This is exact inclusion-exclusion on the divisor poset. It is the arithmetic carrier for reversing “all factor ancestors contributed.”

boundary Cancellation in one finite divisor fiber does not imply square-root cancellation of the cumulative Mertens function.

source Classical Möbius inversion, independently rederived above.

8.0.6 H.0305 Dirichlet-series multiplication

Theorem · proved-standard

depends: H.0303

statement Where both series are absolutely convergent,

$$\left(\sum_{n \geq 1} \frac{f(n)}{n^s} \right) \left(\sum_{n \geq 1} \frac{g(n)}{n^s} \right) = \sum_{n \geq 1} \frac{(f \star g)(n)}{n^s}.$$

derivation Absolute convergence permits rearrangement:

$$\sum_{a,b \geq 1} \frac{f(a)g(b)}{(ab)^s} = \sum_{n \geq 1} \frac{1}{n^s} \sum_{ab=n} f(a)g(b).$$

The inner sum is Dirichlet convolution.

boundary Outside absolute convergence, rearrangement and analytic continuation require separate theorems.

source Classical Dirichlet-series product identity.

8.0.7 H.0306 Euler product

Theorem · proved-standard

depends: H.0302, H.0305

statement For $\operatorname{Re} s > 1$,

$$\zeta(s) = \sum_{n \geq 1} n^{-s} = \prod_p (1 - p^{-s})^{-1}.$$

derivation Expand each local factor as the convergent geometric series $\sum_{k \geq 0} p^{-ks}$. Multiplying a finite prime set gives one term for every valuation vector supported on that set. Unique factorization makes these terms exactly the integers composed from those primes. Absolute convergence lets the finite products exhaust all primes and integers.

transformations The Euler product is the exact junction between integer succession and independent prime-valuation axes.

boundary The product does not converge as such throughout the critical strip. Analytic continuation is a new receiver, not termwise continuation of every local product.

source Euler; see Bombieri's Clay RH account and standard analytic number theory.

8.0.8 H.0307 Prime-power logarithmic current

Theorem · proved-standard

depends: H.0306

statement For $\operatorname{Re} s > 1$,

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_p \sum_{k \geq 1} \frac{\log p}{p^{ks}} = \sum_{n \geq 1} \frac{\Lambda(n)}{n^s},$$

where $\Lambda(p^k) = \log p$ and $\Lambda(n) = 0$ otherwise.

derivation Take logarithms of the absolutely convergent Euler product:

$$\log \zeta(s) = \sum_p \sum_{k \geq 1} \frac{p^{-ks}}{k}.$$

Differentiate termwise; the factor $-k \log p$ cancels $\frac{1}{k}$.

transformations Primes are primitive multiplicative returns; powers are their repeated traversals. The logarithmic derivative weights each repeat by the primitive length $\log p$.

boundary This identity alone gives no sign or location for analytically continued zeros.

source Classical logarithmic derivative of the Euler product.

8.0.9 H.0308 Theta inversion by Poisson summation

Theorem · proved-standard

depends: H.0207, H.0212, H.0247

statement For $t > 0$, let

$$\theta(t) = \sum_{n \in \mathbb{Z}} \exp(-\pi n^2 t).$$

Then

$$\theta(t) = t^{-\frac{1}{2}} \theta\left(\frac{1}{t}\right).$$

derivation The Fourier transform of $x \mapsto \exp(-\pi t x^2)$ is $\xi \mapsto t^{-\frac{1}{2}} \exp\left(-\pi \frac{\xi^2}{t}\right)$. Poisson summation equates the sums of a Schwartz function and its Fourier transform over $\mathbb{Z}$, yielding the identity.

transformations The factor $t^{-\frac{1}{2}}$ is the one-dimensional half-density Jacobian of reciprocal scale. It is the analytic source of the centered $s \mapsto 1 - s$ symmetry after Mellin transport.

boundary Poisson summation and the Gaussian Fourier transform are substantial imported theorems; this entry gives their exact composition, not a new proof of all Fourier analysis.

source Jacobi theta transformation via Poisson summation.

8.0.10 H.0309 Mellin completion and zeta functional equation

Theorem · proved-standard

depends: H.0249, H.0308

statement For $\operatorname{Re} s > 1$,

$$\pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right) \zeta(s) = \frac{1}{2} \int_0^\infty (\theta(t) - 1) t^{\frac{s}{2}-1} dt.$$

Splitting at 1 and applying H.0308 continues the completed expression meromorphically and yields the entire function

$$\xi(s) = \frac{1}{2} s(s-1) \pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right) \zeta(s)$$

with

$$\xi(s) = \xi(1-s).$$

derivation Integrate each Gaussian term using

$$\int_0^\infty \exp(-\pi n^2 t) t^{\frac{s}{2}-1} dt = (\pi n^2)^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right).$$

Absolute convergence permits summation. Split the theta integral at 1; in $(0, 1)$ substitute $t = \frac{1}{u}$ and use $\theta(t) = t^{-\frac{1}{2}} \theta\left(\frac{1}{t}\right)$. The paired integrals are invariant under $s \mapsto 1 - s$; the elementary endpoint terms are cancelled by $\frac{s(s-1)}{2}$.

transformations Euler prime factors, the Gamma/archimedean factor, and endpoint factors become one completed receiver. The critical seam is the fixed set of the involution $s \mapsto 1 - s$.

boundary The functional equation gives symmetry of zeros about the seam; it does not put every zero on the seam.

source Riemann's theta–Mellin continuation; formula normalized as in NIST DLMF §25.4.

8.0.11 H.0310 Centered completed function and RH

Definition · definition

depends: H.0309

statement Define

$$\mathcal{X}(z) = \xi\left(\frac{1}{2} + z\right).$$

Then $\mathcal{X}$ is even:

$$\mathcal{X}(-z) = \mathcal{X}(z).$$

The Riemann Hypothesis is the proposition

$$\forall z \in \mathbb{C} : \mathcal{X}(z) = 0 \rightarrow \operatorname{Re} z = 0.$$

transformations In the traditional coordinate $s = \frac{1}{2} + z$, this is $\operatorname{Re} s = \frac{1}{2}$ for every nontrivial zero.

boundary The definition is not a surrogate involving square roots, finite zero lists, or a renamed OPEN inhabitant.

source Standard RH statement in centered coordinates.

8.0.12 H.0311 Hadamard zero divisor of the completion

Theorem · proved-standard

depends: H.0309, H.0310

statement The entire function ξ has order one and admits a canonical product over its zeros. In a conventional symmetric ordering,

$$\xi(s) = \xi(0) \prod_{\rho} \left(1 - \frac{s}{\rho}\right),$$

with the convergence factor absorbed by pairing/symmetric summation; equivalently the genus-one Hadamard product carries explicit exponential factors.

derivation Stirling bounds for $\Gamma(\frac{s}{2})$ and polynomial bounds for ζ on vertical strips give order one. Hadamard's factorization theorem then represents an order-one entire function by its zero divisor and an exponential polynomial. The functional symmetry fixes the permitted exponential normalization under symmetric grouping.

transformations Logarithmic differentiation turns the zero product into a sum of resolvents. This is the common entrance to Li, Pick, Stieltjes, and spectral receiver routes.

boundary Grouping and convergence conventions must be explicit. A formal product over zeros without Hadamard control is not a proof object.

source Hadamard factorization applied to Riemann's xi function.

8.0.13 H.0312 Explicit formula as source/zero comparison

Theorem · proved-standard
depends: H.0307, H.0309, H.0311

statement For an admissible test function and its Mellin/Fourier transform, contour displacement of

$$-\frac{\zeta'(s)}{\zeta(s)}$$

equates a weighted sum over prime powers with a sum over nontrivial zeros, together with the Gamma and endpoint terms of the completion.

derivation Start where H.0307 converges absolutely. Multiply by the transform of the test function and integrate on a vertical line. Shift the contour through the completed meromorphic function. The residue theorem records the pole at 1, trivial zeros/Gamma poles, and every nontrivial zero. Mellin inversion returns the prime-power side.

transformations The formula is one exact comparison cell:

prime powers + archimedean terms + endpoints = completed zero divisor.

boundary The exact signs and endpoint conditions depend on the chosen test-space convention. The formula alone does not force the zero terms to have one sign.

source Riemann–Weil explicit formula; see Bombieri's official Clay problem description.

8.0.14 H.0313 Graded arithmetic accessibility current

Definition · definition
depends: H.0213, H.0302

statement Let C_e be an oriented finite or locally finite chain of arithmetic occurrences accessible to receiver incidence e , and let $\nu_e : C_e \rightarrow \mathbb{N}^{P_e}$ be its exact valuation address in a declared prime aperture. Define

$$\mathfrak{A}_{e(X)} = \sum_{\alpha \in \mathbb{N}^{P_e}} [C_e|_{\nu_e^{-1}(\alpha)}] X^\alpha.$$

The coefficient is a chain or signed occurrence population. The degree decomposition by $|\alpha|_1$ distinguishes primitive axes X_p , repeated degree-two traversal X_p^2 , and cross-axis degree-two incidence $X_p X_q$.

transformations An augmentation may return an exact count or generating series only after the occurrence, orientation, multiplicity, boundary, and valuation address have been retained.

boundary This is not a prime probability scalar or universal taxonomy. A growing prime aperture carries unresolved support outside its present axes.

source Laboratory chain-valued synthesis of H.0213 and the valuation atlas supplied by the fundamental theorem of arithmetic.

8.0.15 H.0314 Prime-cut incidence calculus

Theorem · proved-derived
depends: H.0215, H.0302, H.0313

statement In a fixed finite aperture let K_p restrict the current away from multiples of a newly admitted prime p , and put $\Delta_p \mathfrak{A} = K_p \mathfrak{A} - \mathfrak{A}$. Then

$$\Delta_p \mathfrak{A} = - \sum_{p|n} [n] X^{\nu(n)}.$$

For distinct new axes p, q ,

$$\Delta_q \Delta_p \mathfrak{A} = \Delta_p \Delta_q \mathfrak{A} = \sum_{pq|n} [n] X^{\nu(n)}.$$

derivation The first difference is exact chain restriction. Applying the second cut restores with positive sign precisely the doubly removed population. Divisibility cuts commute, so their curvature commutator is zero while their joint Hessian may be nonzero.

transformations The degree-two projection of the mixed difference selects $X_p X_q$. Prime squares instead occupy the valuation layer $v_{p(n)} = 2$ and have face X_p^2 ; they are not a fictitious second admission of p .

boundary Static divisibility cuts commute. Receiver-dependent apertures or transported coefficient laws may have nonzero mixed curvature and must be compared using H.0215.

source Unique factorization and finite inclusion–exclusion.

8.0.16 H.0315 Exact wall current of the interval from 113 to 127

Theorem · computational-witness

depends: H.0314

statement Beginning with odd integers in $[113, 127]$ and admitting 3, 5, 7, 11 successively gives

$$\Delta_3 \mathfrak{A} = -(X_3^2 X_{13} + X_3 X_{41}),$$

$$\Delta_5 \mathfrak{A} = -(X_5 X_{23} + X_5^3),$$

$$\Delta_7 \mathfrak{A} = -X_7 X_{17}, \quad \Delta_{11} \mathfrak{A} = -X_{11}^2.$$

The endpoint survivors are 113 and 127.

derivation The six wall crossings are

$$117 = 3^2 \cdot 13, 123 = 3 \cdot 41, 115 = 5 \cdot 23, 125 = 5^3, 119 = 7 \cdot 17, 121 = 11^2.$$

Therefore the endpoint gap 14 is the augmentation of three mixed semiprime crossings, one prime-square crossing, and two degree-three crossings.

transformations The place sets grow exactly as

$$\{2\}, \{2, 3\}, \{2, 3, 5\}, \{2, 3, 5, 7\}, \{2, 3, 5, 7, 11\}.$$

Each stage is an arithmetic boundary change, not a new absolute number line.

boundary This finite witness exposes what the quotient gap forgets. It does not predict primes or establish a global distribution law.

source `src/soma/observations/prime-spectral-world-01/RESULTS.md`, re-expressed exactly through H.0313–H.0314.

9 Algebraic geometry, families, and cohomological arithmetic

9.0.1 H.0320 Affine locus and coordinate ring duality

Theorem · proved-standard

depends: H.0128, H.0220

statement Over an algebraically closed field k , Hilbert’s Nullstellensatz gives

$$I(V(I)) = \sqrt{I}, \quad V(I(V)) = V.$$

Affine algebraic sets correspond contravariantly to finitely generated reduced k -algebras.

transformations Equations and their solution loci are contravariant receiver faces of one constrained algebraic construction.

boundary The algebraic-closure and finite-generation hypotheses matter; schemes retain nilpotents and arithmetic base information beyond reduced loci.

source Hilbert Nullstellensatz and affine coordinate-ring duality.

9.0.2 H.0321 Prime spectrum and affine scheme

Theorem · proved-standard

depends: H.0120, H.0320

statement $\text{Spec}(A)$ is the set of prime ideals with closed sets $V(I) = \{p : I \subseteq p\}$. A ring map $A \rightarrow B$ induces $\text{Spec}(B) \rightarrow \text{Spec}(A)$ by inverse image, and affine schemes are contravariantly equivalent to commutative rings.

transformations A scheme point carries a prime ideal and local algebraic interior; map direction reverses under the spectrum receiver.

boundary A prime ideal is not merely a coordinate point, and arithmetic spectra are not generally varieties over algebraically closed fields.

source Affine scheme–ring duality.

9.0.3 H.0322 Localization, structure sheaf, and stalk

Theorem · proved-standard

depends: H.0030, H.0321

statement On the basic open $D(f) \subseteq \operatorname{Spec}(A)$,

$$\mathcal{O}(D(f)) = A_f,$$

and the stalk at p is $\mathcal{O}_p = A_p$. Compatible local fractions glue uniquely by the sheaf law.

transformations Localization is an exact local receiver inverting the functions visible as nonzero on the selected region.

boundary Local equality in one stalk need not imply equality on the complete scheme.

source Structure sheaf of an affine scheme.

9.0.4 H.0323 Fiber product and algebraic base change

Theorem · proved-standard

depends: H.0025, H.0125, H.0321

statement For $A \rightarrow B, C$,

$$\operatorname{Spec}(B) \times_{\operatorname{Spec}(A)} \operatorname{Spec}(C) \equiv \operatorname{Spec}(B \otimes_A C).$$

transformations Co-present algebraic incidence is a pullback; its coordinate receiver is the tensor product.

boundary Tensoring can introduce nilpotents, reducibility, or singularity; base change does not preserve every property without hypotheses.

source Affine fiber-product theorem.

9.0.5 H.0324 Projective coordinates and Bézout intersection

Theorem · proved-standard

depends: H.0273, H.0320

statement Projective algebraic loci are cut out by homogeneous ideals. If two plane projective curves over an algebraically closed field have degrees m, n and share no component, then

$$\sum_P I_{P(C,D)} = mn$$

with intersection multiplicity.

transformations Projection may change the visible diagram while homogeneous incidence and total intersection multiplicity remain exact.

boundary Set-theoretic crossing count omits multiplicity and points at infinity.

source Projective coordinate theory and Bézout theorem.

9.0.6 H.0325 Noether normalization and dimension

Theorem · proved-standard

depends: H.0130, H.0320

statement For a finitely generated k -domain A , there are algebraically independent $t_1, \dots, t_d$ such that A is finite over $k[t_1, \dots, t_d]$, and

$$d = \dim A = \operatorname{trdeg}_k \operatorname{Frac}(A).$$

transformations Intrinsic algebraic dimension is the number of independent axes in a finite generic projection, not the number of chosen generators.

boundary Dimension can jump in families, and embedding dimension can exceed Krull dimension at singular points.

source Noether normalization theorem.

9.0.7 H.0326 Zariski tangent and Jacobian seam

Theorem · proved-standard

depends: H.0122, H.0325

statement For $X = V(f_1, \dots, f_r) \subseteq \mathbb{A}^n$,

$$T_x X = \ker J_{f(x)}.$$

At expected codimension, maximal Jacobian rank is equivalent to smoothness; rank loss marks the singular locus.

transformations The Jacobian determines first-order afforded directions; its rank discriminant is a precise seam where local continuation type changes.

boundary A singularity is not inferred from a rendered kink alone. In positive characteristic the naive Jacobian criterion needs its standard hypotheses.

source Zariski tangent-space definition and Jacobian criterion.

9.0.8 H.0327 Divisors and Riemann–Roch on a curve

Theorem · proved-standard
depends: H.0130, H.0322, H.0325

statement On a smooth projective curve, principal divisors have degree zero. For a divisor D and canonical divisor K ,

$$\ell(D) - \ell(K - D) = \deg D + 1 - g.$$

transformations Zeros and poles are oriented valuation incidences; Riemann–Roch balances their degree against the available global sections and genus.

boundary Divisor class, chosen divisor, and section space are different objects.

source Riemann–Roch theorem for smooth projective curves.

9.0.9 H.0328 Normalization of an algebraic locus

Theorem · proved-standard
depends: H.0130, H.0321

statement For a finitely generated domain A over a field, its integral closure $\tilde{A}$ in $\text{Frac}(A)$ is finite over A . The map

$$\text{Spec}(\tilde{A}) \rightarrow \text{Spec}(A)$$

is finite and birational and is an isomorphism over the normal locus.

transformations Normalization separates some singular branch presentations while retaining the function field.

boundary It is not an isomorphism at every singularity and does not preserve the original marked construction.

source Finiteness of normalization for affine varieties.

9.0.10 H.0329 Family, fiber, and discriminant

Definition · definition
depends: H.0018, H.0323, H.0326

statement For a morphism $\pi : X \rightarrow S$, the fiber at s is $X_s = X \times_S \text{Spec}(k(s))$. A discriminant is a declared locus in S where smoothness, rank, multiplicity, or another named fiber invariant fails to continue.

transformations Parameter change moves through a family of interiors; the discriminant is the exact algebraic seam of a selected law class.

boundary Different invariants define different discriminants. “Interesting change” is not a mathematical discriminant until the failed property is named.

source Standard scheme-theoretic fibers and discriminant loci.

9.0.11 H.0330 Étale cohomology, Frobenius trace, and purity

Theorem · proved-standard
depends: H.0226, H.0327

statement For a smooth projective variety X over $\mathbb{F}_q$, the Grothendieck–Lefschetz trace formula and cohomological determinant give

$$Z(X, T) = \prod_i \det(1 - T \text{Frob} \mid H_{\text{et}}^i(X, \mathbb{Q}_l))^{(-1)^{i+1}}.$$

Deligne’s purity theorem places Frobenius eigenvalues on H^i at complex modulus $q^{\frac{i}{2}}$.

transformations Point counts, closed arithmetic orbits, cohomological degrees, and the zeta spectrum are exact connected receivers. Purity fixes the half-weight seam.

boundary The classical-number-field zeta function has no accepted object carrying this entire Frobenius/cohomology/purity package.

source Grothendieck trace formula and Deligne’s Weil-conjecture proof.

9.0.12 H.0331 Moduli and quotient stack boundary

Definition · definition

depends: H.0027, H.0121, H.0329

statement A moduli functor assigns to each base its family of objects up to the declared isomorphism and their pullbacks. A fine moduli space represents that functor. When automorphisms obstruct representability, a stack retains the groupoid of objects and isomorphisms.

transformations Parameter space and symmetry quotient are inseparable when objects have stabilizers; the stack retains that isotropy.

boundary A coarse moduli point need not determine a universal family or automorphism lineage.

source Standard moduli-functor and algebraic-stack definitions.

10 Transcendence, special functions, and formulation atlases

10.0.1 H.0350 Algebraic and transcendental elements

Definition · definition

depends: H.0130

statement An element α of an extension of F is algebraic over F when some nonzero polynomial in $F[x]$ vanishes at α . Otherwise it is transcendental. Algebraicity is therefore relative to the base field.

transformations An algebraic element enters through a finite quotient by its minimal polynomial; a transcendental element admits no such finite algebraic relation over the chosen base.

boundary Transcendental does not mean unconstructible, noncomputable, random, or inaccessible to exact series and integral receivers.

source Standard field theory.

10.0.2 H.0351 Exact exponential and e presentations

Theorem · proved-standard

depends: H.0109, H.0208, H.0212, H.0250

statement The entire function

$$\exp(z) = \sum_{n=0}^{\infty} \frac{z^n}{n!}$$

is the unique solution of $y' = y$, $y(0) = 1$, and obeys $\exp(z + w) = \exp(z) \exp(w)$. Its value

$$e = \exp(1) = \sum_{n=0}^{\infty} \frac{1}{n!} = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n.$$

transformations Differential equation, power series, additive-to-multiplicative homomorphism, and binomial-limit lineage are exact nonidentical presentations of one value.

boundary A finite decimal or truncated sum is a receiver face, not e . Each equality above owes its convergence theorem.

source Classical real and complex analysis of the exponential.

10.0.3 H.0352 Exact pi presentations

Theorem · proved-standard

depends: H.0109, H.0208, H.0240, H.0250

statement Euclidean π has exact connected presentations including

$$\pi = 4 \int_0^1 \frac{1}{1+x^2} dx = 4 \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1},$$

and

$$\frac{\pi}{2} = \prod_{n=1}^{\infty} \frac{(2n)^2}{(2n-1)(2n+1)}.$$

The integral, conditionally convergent series, and Wallis product each carry their own exact hypotheses and proof.

transformations Arc, integral, series, and product are receiver faces related by substitution, termwise integration on admitted regions, and product convergence.

boundary No one formula is the identity of π . The occurrence of a prime or semiprime in one coefficient pattern is presentation-relative until an invariant transport theorem connects it to other presentations.

source Arctangent integral and series; Wallis product.

10.0.4 H.0353 Euler transport between exponential and circular faces

Identity · proved-standard

depends: H.0212, H.0351, H.0352

statement Absolute convergence of the exponential series permits separation into even and odd terms:

$$\exp(i\theta) = \cos \theta + i \sin \theta.$$

Hence

$$\exp(i\pi) + 1 = 0.$$

transformations The imaginary-axis exponential trajectory re-presents additive angle as multiplicative unit-circle phase.

boundary The identity relates declared complex-analytic presentations; it does not make e , π , i , 1, and 0 the same occurrence.

source Euler formula derived from the convergent power series.

10.0.5 H.0354 Lindemann–Weierstrass transcendence

Theorem · proved-standard

depends: H.0350, H.0353

statement If $\alpha_1, \dots, \alpha_n$ are distinct algebraic numbers, then $\exp(\alpha_1), \dots, \exp(\alpha_n)$ are linearly independent over the algebraic numbers. In particular $\exp(\alpha)$ is transcendental for nonzero algebraic α ; therefore e and π are transcendental.

derivation $e = \exp(1)$ gives the first consequence. If π were algebraic, then $i\pi$ would be nonzero algebraic while $\exp(i\pi) = -1$ is algebraic, contradicting the theorem.

boundary The theorem does not classify all algebraic relations among values of exponential and logarithmic functions.

source Lindemann–Weierstrass theorem.

10.0.6 H.0355 Gamma and beta continuation

Theorem · proved-standard

depends: H.0249, H.0283, H.0352

statement For $\operatorname{Re}(s) > 0$,

$$\Gamma(s) = \int_0^\infty t^{s-1} \exp(-t) dt, \quad \Gamma(s+1) = s\Gamma(s), \quad \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$$

The beta integral obeys

$$B(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}.$$

transformations Mellin transport, recursion, Gaussian geometry, and simplex integration connect these exact faces.

boundary Meromorphic continuation crosses poles and requires the continued function, not the original integral outside its domain.

source Euler gamma and beta identities.

10.0.7 H.0356 Hypergeometric equation and transformation atlas

Definition · definition

depends: H.0109, H.0252, H.0283

statement For $|z| < 1$ and c not a nonpositive integer,

$${}_2F_1(a, b; c; z) = \sum_{n=0}^{\infty} \frac{(a)_n (b)_n}{(c)_n n!} z^n$$

and it solves

$$z(1-z)y'' + [c - (a+b+1)z]y' - aby = 0.$$

Analytic continuation and parameter transformations relate local solution bases around its singular points $0, 1, \infty$.

transformations Coefficient recurrence, differential equation, singularity monodromy, and transformed series are exact faces of one local system.

boundary A formal parameter substitution is not an identity unless its domain, branch, and continuation path are specified.

source Gauss hypergeometric function theory.

10.0.8 H.0357 Elliptic integral and arithmetic–geometric mean

Theorem · proved-standard

depends: H.0250, H.0356

statement The complete elliptic integral

$$K(k) = \int_0^{\frac{\pi}{2}} \frac{d\theta}{\sqrt{1-k^2 \sin^2 \theta}} = \frac{\pi}{2} {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; k^2\right)$$

satisfies Gauss’s exact identity

$$K(k) = \frac{\pi}{2 \operatorname{AGM}\left(1, \sqrt{1-k^2}\right)}.$$

transformations A period integral, hypergeometric series, and quadratically convergent mean iteration are exact connected presentations.

boundary Branches and parameter domain are part of the identity. Fast convergence does not erase the iteration lineage.

source Gauss hypergeometric and AGM theorem for elliptic integrals.

10.0.9 H.0358 Modular forms and q-expansion

Definition · definition

depends: H.0132, H.0284

statement A modular form of weight k for a declared subgroup Γ obeys

$$f\left(\frac{az+b}{cz+d}\right) = (cz+d)^k f(z)$$

and the required holomorphy at cusps. If the width of the cusp ∞ is h , its exact local coordinate is $q = \exp(2\pi i \frac{z}{h})$ and $f(z) = \sum_{n \geq 0} a_n q^n$.

transformations Upper-half-plane geometry, group action, and coefficient lineage are connected through the cusp receiver.

boundary Weight, level, multiplier, cusp conditions, and coefficient field must be declared; not every q -series is modular.

source Standard modular-form definition and q-expansion principle.

10.0.10 H.0359 Hecke eigenform Euler product

Theorem · proved-standard

depends: H.0305, H.0306, H.0358

statement A normalized cuspidal simultaneous Hecke eigenform of integral weight k , level N , and declared character has multiplicative Fourier coefficients and an L -series

$$L(f, s) = \sum_{n \geq 1} \frac{a_n}{n^s}$$

with local Euler factors determined by its Hecke eigenvalues. Its completed L -function, with Gamma, conductor, and character factors fixed by (k, N) and the chosen normalization, extends holomorphically and obeys the corresponding $s \mapsto k-s$ functional equation.

transformations Symmetry eigenchannels, coefficient recurrence, prime-local factors, and global analytic completion are exact related receivers.

boundary Euler product and functional equation are not automatic for an arbitrary Dirichlet series.

source Hecke theory of modular eigenforms.

10.0.11 H.0360 Continued-fraction path and determinant

Theorem · proved-standard

depends: H.0148, H.0223

statement For a simple continued fraction with convergents $\frac{p_n}{q_n}$,

$$p_n q_{n-1} - p_{n-1} q_n = (-1)^{n-1}.$$

Infinite simple continued fractions converge, and irrational real numbers have unique infinite expansions under the conventional terminal rule.

transformations Each partial quotient selects a left or right fractional-linear step; the determinant retains neighboring orientation exactly.

boundary A convergent is a rational presentation with an exact remainder relation, not the irrational value itself.

source Classical continued-fraction theory.

10.0.12 H.0361 Period number

Definition · definition

depends: H.0240, H.0320, H.0350

statement A period is a complex number whose real and imaginary parts are absolutely convergent integrals of rational functions with algebraic coefficients over semialgebraic domains defined over the algebraic numbers.

transformations Algebraic data, domain incidence, and exact integration form a finite presentation class containing algebraic numbers and π .

boundary It is open whether familiar constants such as e belong to this class. “Transcendental” and “period” are nonidentical classifications.

source Kontsevich–Zagier definition of periods.

10.0.13 H.0362 Transcendental formulation atlas

Definition · definition

depends: H.0023, H.0100, H.0351, H.0352, H.0355, H.0356, H.0357, H.0360, H.0361

statement A formulation node is

$$(D, E, \mathcal{H}, v, \rho),$$

where D is the parameter domain, E an exact expression or algorithm, $\mathcal{H}$ its hypotheses and branch data, v the denoted invariant, and ρ the receiver. An atlas edge is a proved transformation carrying one node to another while preserving v .

transformations Substitution with Jacobian, analytic continuation, functional equation, recurrence, hypergeometric transformation, modular action, and exact algorithm conjugacy are distinct edge species.

boundary Shared output does not supply an edge. The atlas is not complete merely because infinitely many formulas exist.

source Laboratory pi/e/transcendental synthesis in standard atlas language.

10.0.14 H.0363 Prime-valuation receiver of a formulation

Definition · proved-derived

depends: H.0302, H.0362

statement For a formulation whose rational coefficients are $\frac{a_n}{b_n}$ in lowest terms, choose a finite coefficient-index set I and finite prime set P . Its prime-valuation face is

$$\nu_{P,I}(E) = \left(v_{p(a_n)}, v_{p(b_n)} \right)_{(p,n) \in P \times I}.$$

Transformation edges act on this decorated coefficient lineage according to their exact algebraic identities.

transformations This receiver can compare where primes, prime powers, and semiprimes enter coefficient numerators, denominators, recurrences, or modular levels.

boundary The face is presentation-relative. No theorem here says that all formulas for one transcendental invariant share one prime distribution or that a local coefficient pattern determines zeta zeros.

source Laboratory formulation-atlas receiver; valuation theory standard.

10.0.15 H.0364 Transcendental evaluation fibers carry changing arithmetic currents

Theorem · proved-derived

depends: H.0313, H.0351, H.0352, H.0362, H.0363

statement For an exact evaluation map

$$\text{ev} : \mathcal{F} \rightarrow \mathcal{V},$$

the fiber $\mathcal{E}_\tau = \text{ev}^{-1}(\tau)$ contains nonidentical formulations returning τ . A proved transport $T : F \rightarrow G$ has arithmetic residual

$$\text{Res}_T = T_* \mathfrak{A}_F - \mathfrak{A}_G.$$

It vanishes only when that transformation preserves the declared decorated accessibility current.

Valuation specialization obeys

$$\prod_p p^{-s\alpha_p} = \exp \left(-s \sum_p \alpha_p \log p \right),$$

while the factorial lineage of e obeys

$$v_{p((n+1)!)} - v_{p(n!)} = v_{p(n+1)}.$$

derivation For a real rational $x = \frac{u}{v}$ in lowest terms with $v \neq 0$ and $|u| \leq |v|$, let

$$A(x) = \sum_{n \geq 0} (-1)^n \frac{x^{2n+1}}{2n+1}$$

be the exact arctangent series, conditionally at the endpoints. Then

$$\frac{a_{n+1}}{a_n} = \frac{u^2(2n+1)}{v^2(2n+3)}.$$

Thus every p dividing v meets the moving odd current on exact sheets $2n+1 \equiv 0 \pmod{p^k}$ or $2n+3 \equiv 0 \pmod{p^k}$. In the first twelve transitions of Machin's $\pi = 16A(\frac{1}{5}) - 4A(\frac{1}{239})$, the first arm crosses two 5-sheets while the second crosses no 239-sheet. Its Gaussian closures are nevertheless

$$1 + 5^2 = 2 \cdot 13, \quad 1 + 239^2 = 2 \cdot 13^4.$$

transformations Parameter axes 5, 239, Gaussian closure axes 2, 13, and the fixed Chudnovsky scale axes

$$640320 = 2^6 \cdot 3 \cdot 5 \cdot 23 \cdot 29$$

are related but nonidentical receiver faces of exact π formulations.

boundary No invariant has one universal coefficient support. The twelve-step collision result is bounded but exact. These arithmetic/formulation currents supply a direction for completed zeta transport; they do not determine the nontrivial zero set.

source Standard exponential, factorial, arctangent, and Gaussian identities; exact bounded evidence in src/soma/observations/eros-transcendental-presentation-topology-01/RESULTS.md.

11 Computation, information, optimization, and formal dynamics

11.0.1 H.0400 Formal language and automaton

Definition · definition
depends: H.0100, H.0140

statement A formal language is a subset of finite words over an alphabet. A finite automaton is a finite state transition system with initial and accepting states; it recognizes exactly the words whose induced path ends in an accepting state.

transformations Alphabet symbols, state path, and acceptance are distinct receiver layers. Regular languages are closed under Boolean operations and inverse homomorphism.

boundary A recognized word does not reconstruct its state path in a nondeterministic automaton, and a language is not a semantic ontology.

source Classical automata theory.

11.0.2 H.0401 Turing computation and partial function

Definition · definition
depends: H.0103, H.0400

statement A Turing machine has a finite control and an unbounded finite-support tape. Its transition relation generates configurations. It computes a partial function when every accepted input has a unique halting output and divergence denotes undefinedness.

transformations The partial input/output function is one receiver of the complete configuration path.

boundary The model does not prescribe physical hardware, energy, representation cost, interaction with an open world, or semantic meaning.

source Turing-machine definition.

11.0.3 H.0402 Church–Turing thesis boundary

Boundary · project-postulate
depends: H.0110, H.0401

statement The Church–Turing thesis identifies effectively calculable number functions with Turing-computable functions. Equivalent formal models include lambda-definability and partial recursion.

transformations Simulation theorems relate formal machine presentations. The thesis relates those formal classes to the preformal notion of effective method.

boundary It is not a theorem derived from holonics and does not claim that every physical, interactive, stochastic, or semantic process is one closed function.

source Church–Turing thesis and classical equivalence theorems.

11.0.4 H.0403 Decidability, enumeration, and the halting boundary

Theorem · proved-standard

depends: H.0401

statement A set is decidable when a total machine returns its characteristic function and computably enumerable when a partial machine halts exactly on its members. The halting set is computably enumerable but not decidable.

transformations Diagonalization constructs an exact open boundary for a universal machine's self-prediction receiver.

boundary Undecidable does not mean random, physically impossible, or unsolvable for every bounded instance.

source Turing undecidability theorem.

11.0.5 H.0404 Reduction and complexity class

Definition · definition

depends: H.0401

statement A polynomial-time many-one reduction $A \leq_p B$ is a polynomial-time computable f with $x \in A$ exactly when $f(x) \in B$. P contains polynomial-time decidable languages; NP contains languages with polynomially checkable polynomial-size witnesses.

transformations A reduction transports instances and yes/no consequences while retaining a resource bound.

boundary $P = NP$ is open. "Localized $P=NP$," starvation, analogy, or one fast instance is not the standard proposition.

source Classical computational complexity theory.

11.0.6 H.0405 Rewrite relation, termination, and confluence

Theorem · proved-standard

depends: H.0022, H.0021

statement A rewrite system is a relation $\rightarrow$ on terms. It is terminating when no infinite rewrite path exists and confluent when every fork $a \rightarrow b, c$ has a join. Termination plus local confluence implies confluence.

transformations A commuting diamond is an exact join of two causal rewrite paths; a failed diamond is OPEN at that law grain.

boundary Confluence does not imply termination, efficiency, semantic correctness, or equality of the interior paths.

source Newman's lemma and Church–Rosser theory.

11.0.7 H.0406 Lambda calculus and Curry–Howard

Theorem · proved-standard

depends: H.0020, H.0405

statement Simply typed lambda terms obey beta reduction and subject reduction. Under Curry–Howard, propositions correspond to types and normal proof terms to derivations in the matching intuitionistic logic.

transformations Substitution is computational transport; normalization presents a proof or program through another exact syntax path.

boundary The laboratory symbol Lambda names a parameterized transition family, not automatically the lambda calculus. Curry–Howard depends on the chosen logic and type theory.

source Simply typed lambda calculus and Curry–Howard correspondence.

11.0.8 H.0407 Directed completeness and recursive fixed point

Theorem · proved-standard

depends: H.0140, H.0406

statement In a pointed directed-complete partial order, a Scott-continuous $F : D \rightarrow D$ has least fixed point

$$\text{lfp}(F) = \sup_n F^n(\perp).$$

transformations Finite unfoldings form a directed lineage whose supremum receives a recursive computation, including partial or infinite behavior.

boundary The theorem requires directed completeness and Scott continuity; not every recursive system has this domain model.

source Kleene fixed-point theorem in domain theory.

11.0.9 H.0408 Typed hypergraph and double-pushout rewrite

Definition · definition

depends: H.0025, H.0294

statement A typed hypergraph has vertices and hyperedges incident to ordered or typed vertex families. A double-pushout rewrite presents a rule as a span $L \leftarrow K \rightarrow R$, matches L into a host, removes only the complement permitted by the gluing condition, and adjoins R by pushout.

transformations The interface K is the retained seam between the old and new incidence regions.

boundary A graph-rewrite formalism is not identical to a physical universe, simplicial manifold, or computational algorithm until translations are supplied.

source Algebraic graph transformation by double pushout.

11.0.10 H.0409 Cellular automaton and spacetime diagram

Definition · definition

depends: H.0108, H.0294

statement A cellular automaton on a regular cell complex has a finite local state set, a finite neighborhood, and one translation-compatible local update law applied synchronously. Iteration produces a discrete spacetime diagram.

transformations Global propagation is generated by repeated local incidence. A simplicial or asynchronous generalization must explicitly replace regular cells, neighborhood, clock, and update compatibility.

boundary Scalar state per regular cell and synchronous time are model choices, not universal requirements of discrete dynamics.

source Classical cellular automata; generalized-cell boundary stated explicitly.

11.0.11 H.0410 Kolmogorov description complexity

Theorem · proved-standard

depends: H.0111, H.0401

statement For a fixed universal prefix machine U ,

$$K_{U(x)} = \min\{|p| : U(p) = x\}.$$

For universal prefix machines U, V , there is a constant c with $|K_{U(x)} - K_{V(x)}| \leq c$ for every x .

transformations Description length is receiver-relative to a universal interpreter class; invariance is additive, not identity of programs.

boundary K is uncomputable in general and does not retain execution path, runtime, meaning, or causal provenance.

source Invariance theorem for prefix Kolmogorov complexity.

11.0.12 H.0411 Entropy, conditional information, and mutual information

Definition · definition

depends: H.0019, H.0240

statement For a finite distribution p and one fixed logarithm base,

$$H(X) = - \sum_x p(x) \log p(x),$$

with $0 \log 0 = 0$. Conditional entropy and mutual information obey

$$I(X; Y) = H(X) - H(X|Y) = D_{\text{KL}}(p_{X,Y} \text{ norm } p_X p_Y) \geq 0.$$

transformations Entropy is a receiver of a declared recurrence law; mutual information is the residual from product factorization.

boundary These scalars do not identify occurrences, semantic meaning, topology, or the causal path that produced the distribution.

source Shannon entropy and Gibbs inequality.

11.0.13 H.0412 Data-processing inequality

Theorem · proved-standard

depends: H.0411

statement For a Markov chain $X \rightarrow Y \rightarrow Z$,

$$I(X; Z) \leq I(X; Y).$$

More generally relative entropy contracts under a stochastic kernel.

transformations A downstream receiver cannot increase the distinctions measured by this information divergence without new incidence.

boundary The inequality is relative to the selected distributions and divergence. It does not forbid a receiver from making a task-specific feature more accessible.

source Data-processing inequality.

11.0.14 H.0413 Source and channel coding boundary

Theorem · proved-standard

depends: H.0411, H.0412

statement For an independent identically distributed finite-alphabet source, entropy is the sharp asymptotic lossless coding rate. For a declared finite-alphabet memoryless channel, capacity is

$$C = \max_{p(x)} I(X; Y),$$

with reliable rates below C and converse obstruction above C .

transformations Encoding, channel transport, and decoding are a complete declared communication relation; capacity is its asymptotic rate receiver.

boundary Coding theorems do not define semantic meaning or claim that every ecological source is independent, stationary, or memoryless.

source Shannon source and noisy-channel coding theorems.

11.0.15 H.0414 Stochastic kernel and Markov transport

Definition · definition

depends: H.0019, H.0240

statement A stochastic kernel $K(x, dy)$ assigns a probability measure to each source occurrence measurably. Composition is

$$(LK)(x, A) = \int L(y, A)K(x, dy).$$

A stationary measure satisfies $\pi K = \pi$.

transformations Kernels are receiver-relative conditional transport laws; their composition is exact integration over the intermediate occurrence.

boundary A deterministic law is a Dirac-kernel specialization. Stationarity is not equilibrium, reversibility, or ergodicity without further conditions.

source Standard Markov-kernel theory.

11.0.16 H.0415 Convex duality and separating receiver

Theorem · proved-standard

depends: H.0140, H.0242

statement In a locally convex Hausdorff real topological vector space, a nonempty closed convex set and an exterior point can be strictly separated by a continuous linear functional. For proper lower-semicontinuous convex f, g and a continuous linear A , Fenchel–Rockafellar duality relates

$$\inf_x (f(x) + g(Ax))$$

to its conjugate dual; equality holds, for example, when some $x_0 \in \text{dom } f$ has g continuous at Ax_0 .

transformations Convex geometry turns an infeasible incidence into a linear witness and relates primal and dual objective faces.

boundary Dual equality, attainment, uniqueness, and algorithmic efficiency each require separate hypotheses.

source Hahn–Banach separation and Fenchel–Rockafellar duality.

11.0.17 H.0416 Exact computation and certificate

Boundary · proved-derived

depends: H.0005, H.0251, H.0401

statement A bounded computation may carry integers, rationals, algebraic numbers, symbolic expressions, formal series coefficients, or certified rational enclosures. Every emitted claim must be an exact identity, a checked certificate, or a completely quantified relation to one of these carriers.

transformations Exact-arithmetic algorithms may emit progressively refined certified faces without collapsing the denoted construction.

boundary A finite display is testimony only. A certificate must still be checked against its algorithm and hypotheses.

source Computable analysis and exact-real arithmetic boundary.

11.0.18 H.0417 Unitary circuit and measurement receiver

Definition · definition

depends: H.0029, H.0241

statement A finite quantum circuit is a composition and tensoring of unitary maps on finite-dimensional complex Hilbert spaces. A measurement is a positive operator-valued measure whose Born probabilities are

$$p(i) = \langle \psi, E_i \psi \rangle, \quad \sum_i E_i = I.$$

transformations State evolution, subsystem tensor structure, and measurement distribution are distinct mathematical receivers.

boundary This formalism does not make every probabilistic, parallel, or information-geometric algorithm physically quantum.

source Finite-dimensional quantum information theory.

11.0.19 H.0418 Statistical learning problem

Definition · definition

depends: H.0019, H.0411, H.0415

statement A statistical learning problem declares a hypothesis family $\mathcal{H}$, a data law on occurrences, a loss receiver $\ell(h, z)$, and population risk $R(h) = \mathbb{E}[\ell(h, Z)]$. Generalization bounds relate empirical and population receivers through a capacity measure and sampling law.

transformations Parameters, data ecology, prediction face, and loss comparison are separately typed. Optimization selects a member under the declared risk; it does not define truth.

boundary No universal learning law follows from this portal. Model class, ecology, receiver, dependence, and objective determine the theorem.

source Statistical learning theory; laboratory loss regrade.

11.0.20 H.0419 Formal provability, finite witnesses, and incompleteness

Boundary · proved-standard

depends: H.0106, H.0401, H.0403

statement Let T be an effectively axiomatized formal theory containing enough arithmetic to verify finite computations. For a decidable predicate $R(n)$, the universal sentence

$$\Phi = \forall n \in \mathbb{N}, R(n)$$

has Π_1^0 form. Its negation has a finite witness:

$$\neg \Phi \leftrightarrow \exists n \in \mathbb{N}, \neg R(n).$$

Equivalently, a machine which enumerates n and halts at the first failed check halts exactly when Φ is false.

Gödel incompleteness says that every consistent, effectively axiomatized, sufficiently expressive T leaves some arithmetic sentence undecided. It does not say that any named open sentence is independent of T .

transformations The same object has three typed faces: a universal arithmetic assertion, a nonhalting assertion, and the absence of a finite counterexample certificate. If Φ is false, its standard finite witness can be checked inside an arithmetically adequate T . If Φ is true, a finite proof may still establish its universal invariant; otherwise independence from T is a separate metatheorem about the proof system.

boundary “Unresolvable” is not an absolute third truth value. Independence must name T and prove both T does not prove Φ and T does not prove $\neg \Phi$, ordinarily under a stated consistency or soundness assumption. Long computation, emergent complexity, and quantification over infinitely many integers do not by themselves prove independence.

source Gödel incompleteness and computability theory; finite-witness classification of Π_1^0 sentences.

12 Mathematical-physics interfaces

12.0.1 H.0450 Dimensioned quantity and unit transport

Definition · definition

depends: H.0020, H.0200

statement A dimensioned quantity is a member of a one-dimensional torsor or vector line associated with a dimension type. Choosing a unit gives a scalar coordinate; changing unit rescales that coordinate inversely while retaining the quantity.

transformations Tensoring dimension lines composes dimensions. A dimensionless ratio is a scalar receiver only after numerator and denominator have been transported into dual compatible lines.

boundary Dimensional consistency is necessary but not sufficient for a physical law. Equal unit dimensions do not identify source entities or paths.

source Quantity calculus and dimensional analysis in line-space form.

12.0.2 H.0451 Variational action and Euler–Lagrange transport

Theorem · proved-standard

depends: H.0208, H.0252

statement A twice differentiable stationary path for

$$S[q] = \int_{t_0}^{t_1} L(q, \dot{q}, t) dt$$

under fixed-endpoint variations satisfies

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = 0.$$

transformations The action receiver compares neighboring complete paths; integration by parts transports the variation from velocity to position and exposes the boundary term.

boundary Stationary does not mean globally minimal. The action, admissible variations, regularity, and boundary conditions are required data.

source Euler–Lagrange theorem.

12.0.3 H.0452 Noether symmetry and conserved current

Theorem · proved-standard

depends: H.0121, H.0272, H.0451

statement If a differentiable one-parameter transformation leaves an action invariant up to a boundary term, then its Euler–Lagrange solutions carry a conserved Noether current. In field form,

$$\partial_\mu J^\mu = 0$$

on solutions.

transformations Continuous symmetry, variational identity, local current, and integrated charge are exact linked receivers.

boundary Conservation is conditional on the declared action, symmetry, equations of motion, and boundary behavior. It is not derived from $\partial^2 = 0$ alone.

source Noether’s first theorem.

12.0.4 H.0453 Continuity equation across orders of time

Theorem · proved-standard

depends: H.0208, H.0272

statement If density ρ and flux J obey

$$\partial_t \rho + \operatorname{div} J = s,$$

then for a fixed region Ω ,

$$\frac{d}{dt} \int_{\Omega} \rho dV = - \int_{\partial\Omega} J \cdot n dA + \int_{\Omega} s dV.$$

transformations Local change, boundary transport, and source are three faces of one exact balance. Storage inside a region may vary instantaneously while the complete interval balance remains conserved when $s = 0$.

boundary A nonzero source or unaccounted boundary flux is not forbidden by topology; it must be included in the model.

source Divergence theorem applied to the continuity equation.

12.0.5 H.0454 Circuit chains, Kirchhoff current, and voltage

Theorem · proved-standard

depends: H.0205, H.0144, H.0453

statement On an oriented circuit graph, a source-free branch current i satisfies

$$Bi = 0$$

at junctions. A node potential v induces branch voltage $e = B^*v$, and every cycle $z \in \ker B$ obeys

$$\langle e, z \rangle = 0.$$

transformations Junction balance is a chain boundary law; voltage closure is exact cochain evaluation on cycles. Capacitors and inductors add state and constitutive time laws rather than violating these incidence identities.

boundary Kirchhoff laws do not determine resistance, capacitance, inductance, source behavior, transients, or electromagnetic radiation without the corresponding constitutive and field model.

source Algebraic circuit theory.

12.0.6 H.0455 Stress as momentum-flux tensor

Definition · definition

depends: H.0125, H.0270, H.0453

statement In continuum mechanics, Cauchy stress σ is the linear map whose traction on an oriented surface normal n is $t(n) = \sigma n$. Momentum balance is

$$\rho \dot{v} = \operatorname{div} \sigma + \rho b.$$

transformations Stress is transport of momentum across oriented cuts; pressure, shear, and traction are receiver faces of the tensor.

boundary Stress is not heat, scalar loss, or information by definition. Constitutive laws determine how material deformation relates to σ .

source Cauchy stress theorem and continuum momentum balance.

12.0.7 H.0456 Thermodynamic state differential and heat boundary deed

Definition · definition

depends: H.0208, H.0450

statement Internal energy U is a state function. For a process,

$$dU = \delta Q + \delta W_{\text{on}},$$

where heat and work are path-dependent energy transfers. In a simple reversible system,

$$dU = T dS - p dV + \sum_i \mu_i dN_i.$$

transformations State difference and boundary transfer are nonidentical receivers of one process lineage.

boundary Heat is not a state substance or intrinsic punishment. The differential form and sign convention depend on the declared thermodynamic system.

source First law and fundamental thermodynamic relation.

12.0.8 H.0457 Canonical ensemble and free-energy relation

Theorem · proved-standard

depends: H.0019, H.0351, H.0411, H.0456

statement For discrete finite energies E_i at inverse temperature $\beta > 0$,

$$Z(\beta) = \sum_i \exp(-\beta E_i), \quad p_i = \frac{\exp(-\beta E_i)}{Z(\beta)}, \quad F = -\beta^{-1} \log Z.$$

The identity

$$\mathcal{F}_{\beta(q)} = \sum_i q_i E_i + \beta^{-1} \sum_i q_i \log q_i, \quad D_{\text{KL}}(q \text{ norm } p) = \beta (\mathcal{F}_{\beta(q)} - F)$$

for every probability vector q , with $0 \log 0 = 0$. Thus relative entropy and nonequilibrium free energy are related exactly for this declared ensemble.

transformations Energy spectrum, Gibbs recurrence law, partition function, and free energy are exact connected receivers in one model.

boundary This does not identify every loss with free energy or every probability field with thermodynamics.

source Canonical ensemble and Gibbs variational principle.

12.0.9 H.0458 Maxwell field in exterior form

Theorem · proved-standard

depends: H.0271, H.0272, H.0450

statement On an oriented Lorentzian spacetime, in a fixed unit and sign convention, electromagnetic field strength F and current 3-form J obey

$$dF = 0, \quad d \star F = J.$$

Hence $dJ = 0$.

transformations Field, dual metric receiver, source current, flux, and charge are related through exact differential and boundary laws.

boundary $d^2 = 0$ gives charge-current consistency after Maxwell's sourced law is supplied; it does not derive the law, metric, units, or material response.

source Maxwell equations in differential-form notation.

12.0.10 H.0459 Lorentzian metric, causal cone, and proper time

Definition · definition

depends: H.0270, H.0274

statement A Lorentzian metric has signature $(-, +, \dots, +)$ up to convention and has a unique metric-compatible torsion-free connection. Nonzero tangent vectors are timelike, null, or spacelike according to the sign of $g(v, v)$. Timelike worldlines carry proper time

$$\tau = \int \sqrt{-g(\dot{\gamma}, \dot{\gamma})} d\lambda$$

in units with $c = 1$.

transformations The light cone is the receiver-relative boundary between causal tangent species determined by the metric.

boundary A generic graph cone, algorithmic affordance cone, or visual camera frustum is not a physical light cone without a Lorentzian metric and propagation law.

source Lorentzian geometry.

12.0.11 H.0460 Einstein curvature–source equation

Theorem · proved-standard

depends: H.0276, H.0455, H.0459

statement General relativity posits

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu},$$

where Λ is constant and $G_{\mu\nu} = \text{Ric}_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}$. The contracted Bianchi identity gives $\nabla^\mu G_{\mu\nu} = 0$, so a solution owes $\nabla^\mu T_{\mu\nu} = 0$.

transformations Metric curvature and stress–energy flux are linked by a declared dynamical field equation; covariance transports it between charts.

boundary Holonic geometry does not derive the physical constants, Lorentzian signature, matter coupling, or Einstein equation merely from analogy. Replacing π by an informal arc symbol is not this equation.

source Einstein field equation and contracted Bianchi identity.

12.0.12 H.0461 Quantum state, observable, charge, spin, and mass

Definition · definition

depends: H.0121, H.0243, H.0417, H.0459

statement A quantum state is a positive normalized functional, or density operator in a Hilbert representation. Observables are self-adjoint operators. Charge labels a representation of a gauge symmetry; spin labels an irreducible representation of the rotation or Lorentz cover; mass labels the Casimir invariant $P_\mu P^\mu$ in a relativistic representation.

transformations State, measurement distribution, gauge sector, rotational holonomy, and relativistic representation are distinct connected mathematical faces.

boundary Charge, spin, and mass are not derived merely from triangle orientation, polarity, or gyration. A specific symmetry group and representation law are required.

source Quantum mechanics and Wigner representation classification.

12.0.13 H.0462 Local observable algebra in quantum field theory

Definition · definition

depends: H.0023, H.0243, H.0459, H.0461

statement Algebraic quantum field theory assigns an observable algebra $\mathcal{A}(O)$ to each admissible spacetime region O , covariantly under inclusion, with spacelike separated algebras commuting under locality.

transformations Region, inclusion, causal separation, observable algebra, and state are an exact local-to-global network rather than one universal scalar field.

boundary The Haag–Kastler axioms are a physical mathematical framework, not proof that every information ecology is quantum field theory.

source Haag–Kastler algebraic QFT axioms.

12.0.14 H.0463 Physical law and holonic translation

Boundary · proved-derived

depends: H.0031, H.0452, H.0458, H.0460, H.0462

statement A physical theory may be imported as a presented model: state spaces, observables, units, geometric structures, evolution equations, symmetry, boundary conditions, and empirical receivers. A holonic translation must map each named structure and prove preservation of the selected laws.

transformations Shared category, topology, differential, conservation, or symmetry carriers explain rigorous analogies between domains without identifying their material interpretations.

boundary No metaphor about heat, lightning, gravity, quantum collapse, or entanglement substitutes for this translation. Conversely, physical mathematics remains a valid source of exact reusable structure.

source Laboratory physics regrade under the theory-translation portal H.0031.

12.0.15 H.0464 Six-shell, receiver frame, and symmetric-stress counts

Atlas · proved-derived

depends: H.0241, H.0270, H.0294, H.0455, H.0459

statement Several exact occurrences of six must remain typed apart:

- a triangle boundary has three vertex incidences and three edge incidences, while adjoining its two-dimensional interior adds another cell;
- an n -cube has $2n$ codimension-one facets;
- a symmetric bilinear form on an n -dimensional space has $\frac{n(n+1)}{2}$ independent components; and
- an affine orthogonal frame group in dimension n has

$$n + \binom{n}{2} = \frac{n(n+1)}{2}$$

translation-plus-rotation generators.

The facet count and symmetric-tensor count agree for positive n exactly when $n = 3$. In $3 + 1$ spacetime dimensions a symmetric stress–energy tensor has ten components, split relative to a timelike receiver as

$$10 = 1 + 3 + 6,$$

namely energy density, momentum/energy flux, and spatial symmetric stress.

transformations Receiver choice turns one covariant tensor into density, current, and stress faces. Changing receiver recomposes those faces without changing the tensor. The numerical coincidences above can therefore guide a translation only after the incidence or representation map is supplied.

boundary Six graph hops are not a universal locality horizon. Equal dimensions do not provide a canonical isomorphism, and the ten components of a four-dimensional stress–energy tensor do not cause the ten-dimensional critical superstring result.

source Elementary polytope incidence, Lie-group dimension counting, and relativistic stress–energy decomposition.

12.0.16 H.0465 Variational tension and the curvature obligation of a knot

Theorem · proved-standard

depends: H.0258, H.0271, H.0287, H.0451, H.0455

statement Tension is not a bare crossing count. Given a declared curve energy

$$E[\gamma] = \int L(\gamma(s), \dot{\gamma}(s)) \, ds,$$

its first variation is the force covector on admissible deformations; an equilibrium curve makes that variation vanish subject to its constraints.

For a closed twice-differentiable space curve, Fenchel’s theorem gives total curvature at least 2π . The Fary–Milnor theorem sharpens this for every nontrivial knot:

$$\int |\kappa(s)| \, ds > 4\pi.$$

transformations Projection crossings may change under receiver motion while knot type and the total-curvature obstruction survive the allowed ambient isotopy. A perturbation of a constrained energy minimizer propagates according to the selected elastic or wave law, not according to knot type alone.

boundary Total curvature is not physical string tension, voltage, heat, or stored energy. Each requires an energy functional, constitutive law, units, and boundary conditions. Knot invariance constrains possible embeddings but does not determine their dynamics.

source Fenchel's theorem and Denne and Sullivan, a new proof of the Fary–Milnor theorem .

12.0.17 H.0466 Field-coupled leader growth and later return current

Construction · proved-derived
depends: H.0258, H.0453, H.0458

statement A precise minimal leader model is a moving conductive domain Ω_t coupled to a potential:

$$\operatorname{div}(\kappa_t \operatorname{grad} \varphi_t) = -\rho_t, \quad J_t = -\kappa_t \operatorname{grad} \varphi_t.$$

The declared local growth law extends Ω_t along an admissible boundary deed derived from the contemporary field. That extension changes κ_t and the boundary, so the potential must be solved again on the successor domain. Once a conducting channel joins the relevant boundary regions, a later high-current return can traverse the channel.

transformations The leader is topology-building field response; the return stroke is a later current afforded by the changed topology. Branching, shielding, reconnection, and sideways leaders arise from repeatedly coupling the field solve to the changed domain rather than selecting a path in a fixed graph.

boundary This is a mathematical construction species, not a complete atmospheric lightning simulation. A deterministic instance owes complete initial and boundary data plus an exact growth law. Least action does not imply a greedy shortest-path search.

source Dielectric-breakdown moving-boundary models and measured leader–return-stroke separation in lightning physics.

12.0.18 H.0467 Worldsheet stress closure and critical superstring dimension

Theorem · proved-standard
depends: H.0279, H.0451, H.0455, H.0460, H.0461

statement In the Ramond–Neveu–Schwarz worldsheet formulation, D coordinate bosons and D Majorana fermions contribute matter central charge

$$c_{\text{matter}} = D + \frac{D}{2} = \frac{3D}{2}.$$

Reparametrization and superconformal ghosts contribute $c_{\text{ghost}} = -15$. Quantum Weyl consistency requires $c_{\text{matter}} + c_{\text{ghost}} = 0$, hence the critical spacetime dimension is $D = 10$.

For a nonlinear sigma-model target metric, the leading metric beta function is

$$\beta_{\mu\nu}^G = \alpha' \operatorname{Ric}_{\mu\nu} + O(\alpha'^2)$$

when the other background fields are suppressed. Vanishing worldsheet trace anomaly therefore yields the vacuum Einstein equation at leading order, while the corresponding renormalization flow begins with Ricci flow.

transformations Worldsheet stress conservation and trace closure constrain the dimension and the admissible target geometry. Thus physical spacetime dynamics appears as a consistency condition of a two-dimensional field theory, rather than from counting the components of four-dimensional stress.

boundary Ten is conditional on this superstring worldsheet content and anomaly cancellation. Background antisymmetric tensor and dilaton fields, higher orders in α' , and other string formulations add their declared terms. This is not a universal maximum dimension for every geometry.

source D. Tong, String Theory , worldsheet central charge and sigma-model beta functions.

12.0.19 H.0468 Eleven-dimensional supergravity closure and the Type IIA circle

Theorem · proved-standard
depends: H.0459, H.0460, H.0461, H.0467

statement Under the conventional assumptions of one timelike direction, local supersymmetry with at most 32 real supercharges, and no massless field of spin greater than two, eleven is the maximal supergravity spacetime dimension. Its bosonic fields have

$$44 \text{ graviton modes} + 84 \text{ three-form modes} = 128,$$

matching the 128 on-shell gravitino modes.

Strongly coupled Type IIA string theory opens a circular eleventh direction with

$$R_{11} = g_s \ell_s,$$

and the metric rebase

$$ds_{11}^2 = \exp\left(-2\frac{\varphi}{3}\right) ds_{10}^2 + \exp\left(4\frac{\varphi}{3}\right) (dy + C_1)^2.$$

An M2-brane wrapped around that circle has a $(1 + 1)$ -dimensional worldsheet in ten dimensions; the transverse count is retained as $11 = 3 + 8$ becoming $10 = 2 + 8$.

transformations The coupling controls a geometric radius, so the ten-dimensional and eleven-dimensional descriptions are neighboring faces of one limiting construction. Wrapped and unwrapped objects become different lower-dimensional species under the same circle rebase.

boundary Eleven is a maximum only under the stated supergravity assumptions, not an absolute theorem that no twelve-dimensional or auxiliary formulation can exist. The circle relation is the Type IIA/M-theory duality limit, not a derivation of all physical dimensions from holonic counting.

source Witten, String theory dynamics in various dimensions Samtleben, Eleven-dimensional supergravity .

13 Independent Riemann-Hypothesis route atlas

13.0.1 RH.0000 Riemann Hypothesis

Conjecture · conjecture

depends: H.0310

statement

$$\forall z \in \mathbb{C} : \mathcal{X}(z) = 0 \rightarrow \operatorname{Re} z = 0, \quad \mathcal{X}(z) = \xi\left(\frac{1}{2} + z\right).$$

transformations Every route below must prove both arrows between this proposition and its own criterion. The route is complete only when its source-side criterion is established without assuming the zero locations.

boundary Unresolved. No laboratory entry in this registry proves RH.

source Bombieri, official Clay problem description .

13.0.2 RH.0010 Centered positive-real and Pick-kernel route

Equivalence · proved-standard

depends: H.0126, H.0244, H.0310, H.0311, RH.0000

statement Put

$$F(z) = \frac{\mathcal{X}'(z)}{\mathcal{X}(z)}.$$

Then RH is equivalent to F being holomorphic with $\operatorname{Re} F(z) \geq 0$ on $\operatorname{Re} z > 0$. It is also equivalent to positive semidefiniteness, on that half-plane, of

$$K_{F(p,q)} = \frac{F(p) + \overline{F(q)}}{p + \overline{q}}.$$

derivation Under RH, pair the even zero divisor at $\pm i\gamma$:

$$F(z) = \sum_{\gamma > 0} m_{\gamma} \frac{2z}{z^2 + \gamma^2}.$$

For $z = x + iy$, $x > 0$,

$$\operatorname{Re} \frac{2z}{z^2 + \gamma^2} = \frac{2x(x^2 + y^2 + \gamma^2)}{|z^2 + \gamma^2|^2} > 0.$$

Conversely, holomorphy of $F = \frac{\mathcal{X}'}{\mathcal{X}}$ excludes zeros in the right half-plane; evenness excludes zeros in the left half-plane. For the kernel statement, apply the Cayley transform $S = \frac{F-1}{F+1}$. The Schur kernel of S is a nonzero congruence of K_F ; diagonal positivity recovers $\operatorname{Re} F \geq 0$.

transformations Holonic face: F is a normal logarithmic response; K_F compares two receiver incidences. A proof may enter by constructing a source-native passive/Gram realization.

boundary OPEN: derive $\operatorname{Re} F \geq 0$ or a Gram factorization of K_F from the theta–Euler–Gamma source without first assuming RH. A finite kernel cut or a quotient realization is insufficient.

source Hinkkanen’s logarithmic-derivative criterion, indexed by the American Institute of Mathematics RH equivalent list Hadamard product; classical positive-real/Schur kernel theorem.

13.0.3 RH.0020 Weil explicit-formula positivity

Equivalence · proved-standard

depends: H.0126, H.0312, RH.0000

statement On Weil's admissible test space, the completed explicit-formula quadratic form is nonnegative for every test if and only if RH holds. With a Mellin test g , the spectral face has the form

$$Q(g) = \sum_{\rho} \hat{g}(\rho) \overline{\hat{g}(1-\bar{\rho})},$$

with the precise endpoint constraints and normalization of the chosen convention.

derivation Under RH, $1 - \bar{\rho} = \rho$, so the spectral sum is a sum of squared moduli. Conversely, an off-seam zero and its functional-equation companions permit an admissible test concentrated on that finite spectral configuration, producing a negative direction. The explicit formula transports the same form to prime powers plus archimedean and endpoint terms.

transformations This is the most complete established source/zero comparison cell. It is one route, not the definition of RH or holonics.

boundary OPEN: prove the sign on the complete admissible test space. Positivity on finite support, one test family, one aperture, or an eventual parameter regime does not imply the universal form.

source Bombieri, Clay RH account, explicit formula and Weil criterion .

13.0.4 RH.0030 Li-Keiper recurrence coefficients

Equivalence · proved-standard
depends: H.0311, H.0212, RH.0000

statement Define

$$\lambda_n = \frac{1}{(n-1)!} \left[\frac{d^n}{ds^n} (s^{n-1} \log \xi(s)) \right]_{s=1}.$$

With symmetric zero summation,

$$\lambda_n = \sum_{\rho} \left[1 - \left(1 - \frac{1}{\rho} \right)^n \right].$$

Then

$$\text{RH} \leftrightarrow \forall n \geq 1 : \lambda_n \geq 0.$$

derivation The Möbius rebase $w = 1 - \frac{1}{\rho}$ sends the critical line to the unit circle. Under RH, conjugate pairs contribute

$$2 - 2 \cos(n\theta) \geq 0.$$

Li's converse theorem shows that if every coefficient is nonnegative, no transformed zero can lie off the unit circle; an exterior member would eventually contribute exponentially growing oscillation.

transformations Each λ_n is an n -fold recurrence receiver of the complete zero divisor. This route makes the projective/rebase logic explicit.

boundary OPEN: derive positivity for every order n from the arithmetic source. Finite or asymptotic positivity does not close the criterion.

source X.-J. Li, The Positivity of a Sequence of Numbers and the Riemann Hypothesis .

13.0.5 RH.0040 Squared-seam Stieltjes and connected Hankel route

Equivalence · conditional
depends: H.0126, H.0248, H.0311, RH.0010

statement Since $\mathcal{X}$ is even, define the entire function

$$H(w) = \frac{\mathcal{X}(\sqrt{w})}{\mathcal{X}(0)}.$$

RH is equivalent to all zeros of H lying on the negative real axis. The genus-zero logarithmic product of this order- $\frac{1}{2}$ entire function gives, under RH,

$$\frac{H'(w)}{H(w)} = \int_{[0, \infty)} \frac{d\mu(t)}{w+t} \quad \text{with} \quad \mu = \sum_{\gamma > 0} m_{\gamma} \delta_{\gamma^2}.$$

Expanding at $w = 0$ gives

$$\frac{H'(w)}{H(w)} = \sum_{n \geq 0} (-1)^n \mu_n w^n, \quad \mu_n = \int t^{-n-1} d\mu(t).$$

If the resulting reciprocal moment representation is determinate and its Stieltjes transform analytically continues as $\frac{H'}{H}$, it exists exactly when both Hankel families

$$[\mu_{i+j}]_{i,j=0}^N, \quad [\mu_{i+j+1}]_{i,j=0}^N$$

are positive semidefinite for every N .

derivation Under RH, zeros $\pm i\gamma$ fold to $-\gamma^2$, and

$$\frac{H'}{H} = \sum_{\gamma>0} \frac{m_\gamma}{w + \gamma^2},$$

a positive atomic Stieltjes transform. Conversely, a meromorphic Stieltjes logarithmic derivative has poles only on the negative real axis with positive residues; analytic uniqueness then places the zero divisor of H there. The Stieltjes moment theorem is equivalent to the two connected Hankel positivity families when its existence hypotheses are met.

transformations The fold $w = z^2$ quotients the two seam orientations while retaining squared distance. Overlapping Hankel cells are finite receiver windows of one moment lineage.

boundary OPEN: construct the positive source measure or Gram representation after the nonlinear logarithmic transform. Positive theta moments alone do not imply this. The exact analytic/moment determinacy hypotheses must remain visible.

source Classical Stieltjes moment theorem applied to the centered Hadamard product.

13.0.6 RH.0050 Nyman–Beurling–Báez-Duarte exact closure

Equivalence · proved-standard

depends: H.0250, H.0306, H.0309, RH.0000

statement Let $\rho(x) = x - \lfloor x \rfloor$ and use the standard fractional-part dilation family

$$\rho_{a(x)} = \rho\left(\frac{1}{ax}\right)$$

in $L^2(0, \infty)$. The Báez-Duarte strengthening states that RH is equivalent to the indicator of $(0, 1)$ belonging to the closed linear span of

$$\{\rho_a : a \in \mathbb{N}\}$$

.

derivation Mellin–Plancherel transports each fractional-part dilation to a multiple of $\frac{\zeta(s)}{s}$. Membership of the indicator, whose Mellin transform is $\frac{1}{s}$, in the exact Hilbert-space closure is therefore equivalent to a sequence of finite-span inverse faces converging in norm on the critical boundary. The zero-free half-plane and closure argument give the two directions.

transformations Each dilation is a rebase face of the counting discontinuity. The route asks whether their finite composites converge to a receiver-exact inverse.

boundary OPEN: construct a finite-span sequence whose global L^2 residual converges exactly to zero. Pointwise convergence, a formal Möbius inverse, or a finite dilation window does not establish norm closure.

source L. Báez-Duarte, A strengthening of the Nyman–Beurling criterion .

13.0.7 RH.0060 Laguerre–Pólya and Jensen-polynomial hyperbolicity

Equivalence · proved-standard

depends: H.0109, H.0250, H.0310, RH.0000

statement Let $\gamma(n)$ be the Taylor-coefficient lineage of a standard real entire normalization of Ξ , and define

$$J_\gamma^{d,n}(X) = \sum_{j=0}^d \binom{d}{j} \gamma(n+j) X^j.$$

RH is equivalent to membership of Ξ in the Laguerre–Pólya class, equivalently to hyperbolicity of every $J_\gamma^{d,n}$.

derivation A real entire function is in the Laguerre–Pólya class exactly when it is a locally uniform limit of real-rooted polynomials. Jensen’s theorem converts this property to real-rootedness of every Jensen polynomial. The real zeros of the centered Ξ correspond exactly to RH.

transformations (d, n) is a bifiltration: degree is receiver width; n is lineage shift. Hyperbolicity means every crossing of that finite coefficient face remains on one real orientation.

boundary OPEN: prove hyperbolicity for the complete two-parameter lattice. Hyperbolicity for every fixed degree at sufficiently large shift leaves an unbounded initial region as degree grows.

source Griffin–Ono–Rolen–Zagier, Jensen polynomials for the Riemann zeta function .

13.0.8 RH.0070 de Bruijn–Newman phase flow

Equivalence · proved-standard

depends: H.0018, H.0253, H.0308, H.0310, RH.0000

statement A theta-derived Fourier kernel defines

$$H_{t(z)} = \int_0^\infty \exp(tu^2) \Phi(u) \cos(zu) du.$$

There is a finite constant Λ_{DN} such that H_t has only real zeros exactly for $t \geq \Lambda_{\text{DN}}$. RH is equivalent to $\Lambda_{\text{DN}} \leq 0$. Rodgers and Tao proved $\Lambda_{\text{DN}} \geq 0$, hence RH is equivalent to $\Lambda_{\text{DN}} = 0$.

derivation At $t = 0$, H_0 is a real-variable normalization of the completed zeta function. De Bruijn proves forward preservation above a threshold; Newman proves existence of the sharp threshold. The centered real-zero condition at $t = 0$ is RH. Rodgers–Tao establish the opposite inequality.

transformations t is a true parameterized phase flow. The threshold is a discriminant at which off-axis conjugate zero pairs collide with or leave the real seam.

boundary OPEN: prove real-rootedness at the undeformed face $t = 0$. Positivity of Φ does not by itself imply real zeros of its Fourier transform, and forward regularization does not automatically reverse.

source Rodgers–Tao, The de Bruijn–Newman constant is non-negative .

13.0.9 RH.0080 Hermite–Biehler, de Branges, and canonical systems

Equivalence · proved-standard
depends: H.0243, H.0310, RH.0000

statement For entire E , let $E^\sharp(z) = \overline{E(\bar{z})}$. The Hermite–Biehler inequality

$$|E(z)| > |E^\sharp(z)| \quad (\text{Im } z > 0)$$

forces the real and imaginary companions to have real interlacing zeros. In Suzuki’s xi-derived family,

$$\Theta_{\omega(z)} = \frac{\xi(\frac{1}{2} - \omega - iz)}{\xi(\frac{1}{2} + \omega - iz)},$$

RH is equivalent to meromorphic innerness for every $\omega > 0$.

derivation The quotient $\frac{E^\sharp}{E}$ is Schur/inner exactly when the Hermite–Biehler orientation holds. De Branges theory realizes such a function through a canonical system with a positive-semidefinite Hamiltonian. Suzuki identifies the zero-free half-plane $\text{Re } s > \frac{1}{2} + \omega_0$ with innerness for every $\omega > \omega_0$; setting $\omega_0 = 0$ gives RH.

transformations $E, E^\sharp$ are opposed oriented faces; Θ_ω is their boundary transport; the Hamiltonian is the local metric admitting continuation.

boundary OPEN: construct the xi-derived positive Hamiltonian or innerness for every $\omega > 0$ without using the desired zero-free region. A safe parameter range, quotient, or assumed inner function is not the complete bridge.

source M. Suzuki, A canonical system arising from the Riemann zeta-function .

13.0.10 RH.0090 Speiser critical-point topology

Equivalence · proved-standard
depends: H.0210, H.0283, H.0309, RH.0000

statement RH is equivalent to the absence of nonreal zeros of $\zeta'(s)$ in

$$0 < \text{Re } s < \frac{1}{2}.$$

derivation The argument principle and the functional symmetry relate off-seam zeros of ζ to branch/critical events in its level-curve topology. Speiser’s theorem proves both directions: an off-seam zero forces a derivative zero in the left half-strip, and such a derivative zero forces failure of the critical-line zero geometry.

transformations Zeros of ζ' are genuine inflection/branch events of the holomorphic receiver. This is the standard route closest to the laboratory intuition about kinks, folds, and receiver-visible critical crossings.

boundary OPEN: exclude every such derivative zero. A visualization of some level curves or finite zero window does not supply the global argument.

source Speiser’s theorem; modern statement: Garunkštis–Tamošiūnas .

13.0.11 RH.0100 Hilbert–Pólya spectral determinant

Programme · conjecture
depends: H.0243, H.0311, RH.0000

statement Seek a noncircular Hilbert space and self-adjoint operator A such that the complete spectral determinant satisfies

$$\mathcal{X}(z) = \exp(a + bz) \det_{\star}(zI - A)$$

with every zero and multiplicity represented.

derivation If such an identity and self-adjointness are proved, spectral parameters are real, so zeros of the real-variable Ξ lie on its real axis, equivalently RH. A trace formula for A must reproduce every prime-power, Gamma, and endpoint term in H.0312 to establish the determinant identity.

transformations The spectrum is one receiver; periodic/closed source paths are another. The trace formula is the required comparison cell.

boundary OPEN: construct A , prove self-adjointness, the exact determinant/trace identity, and exhaustivity. Matching zero counts, pair correlation, finitely many ordinates, or only the smooth asymptotic is insufficient.

source Hilbert–Pólya programme; Berry–Keating supplies a counting-asymptotic model, not the complete determinant.

13.0.12 RH.0110 Trace formula and noncommutative geometry

Programme · conditional

depends: H.0312, RH.0100

statement Construct an arithmetic dynamical/noncommutative space whose geometric trace formula has primitive closed returns at lengths $\log p$ and whose spectral side is the complete zero divisor of ξ .

derivation The Selberg trace formula demonstrates this mechanism for zeta functions of hyperbolic surfaces. Connes’s adèle-class construction realizes critical zeros as absorption spectrum and possible off-seam zeros as resonances; an exact positive trace identity excluding those resonances would imply RH.

transformations Primes are primitive closed source worldlines; prime powers are repeated returns; trace is a receiver summing fixed cycles.

boundary OPEN: the complete global trace/positivity theorem with all local factors and no residual off-seam resonance.

source A. Connes, Trace formula in noncommutative geometry and the zeros of zeta .

13.0.13 RH.0120 Function-field cohomology and Weil conjectures

Analogue · proved-standard

depends: H.0312, H.0330

statement For a smooth projective variety X over $\mathbb{F}_q$,

$$Z(X, T) = \prod_i \det(1 - T \text{Frob}_{H_{\text{ét}}^i(X)})^{(-1)^{i+1}}.$$

Deligne’s purity theorem places every Frobenius eigenvalue on H^i at modulus $q^{\frac{i}{2}}$, proving the function-field RH.

derivation Point counts are traces of Frobenius powers by the Lefschetz trace formula. The determinant identity packages those traces into $Z(X, T)$. Poincaré duality pairs reciprocal eigenvalues; Deligne’s weight/purity theorem fixes their modulus, not duality alone.

transformations Frobenius is causal transport; cohomological degree is a typed phase; duality pairs opposed faces; purity fixes the half-weight seam.

boundary No accepted classical-number-field object supplies the required cohomology, Frobenius evolution, determinant identity, duality, and purity. The Weil conjectures are a proven analogue and design constraint, not the standard which every holonic proof must imitate.

source P. Deligne, La conjecture de Weil I .

13.0.14 RH.0130 Prime-counting and Möbius residuals

Equivalence · proved-standard

depends: H.0249, H.0261, H.0304, H.0312, RH.0000

statement Standard equivalent forms include

$$\pi(x) = \text{Li}(x) + O(\sqrt{x} \log x),$$

$$\psi(x) = x + O(\sqrt{x} \log^2 x),$$

and

$$M(x) = \sum_{n \leq x} \mu(n) = O_{\varepsilon}(x^{\frac{1}{2}+\varepsilon})$$

for every $\varepsilon > 0$.

derivation Perron/Mellin inversion writes counting functions as contour integrals of $-\frac{\zeta'}{\zeta}$ or $\frac{1}{\zeta}$. Moving the contour to the seam yields the square-root scale under RH. Conversely, any zero with real part $\beta > \frac{1}{2}$ contributes a residue of scale x^{β} , contradicting the stated bounds.

transformations This is the source residual before the zero receiver. The $\frac{1}{2}$ is the sharp scaling exponent forced by the centered zero boundary, not a base-10 artifact.

boundary OPEN: prove the uniform square-root-scale cancellation. Finite prime patterns, digit rotations, and local residue wheels can contribute exact source identities but do not alone supply the global bound.

source Bombieri, Clay RH account standard Möbius equivalence.

13.0.15 RH.0140 Robin–Lagarias divisor inequalities

Equivalence · proved-standard

depends: H.0302, RH.0000

statement Robin’s criterion is

$\text{RH} \leftrightarrow \sigma(n) < \exp(\gamma)n \log \log n$ for every $n > 5040$.

Lagarias proved the equivalent elementary inequality

$$\sigma(n) \leq H_n + \exp(H_n) \log H_n \quad \text{for every } n \geq 1,$$

with equality only at $n = 1$.

derivation The maximal order of $\frac{\sigma(n)}{n}$ is governed by integers with dense small-prime valuation structure. Robin reduces a possible violation to the colossally abundant regime and relates its asymptotics to the zero-free boundary. Lagarias uses precise bounds comparing H_n with $\log n + \gamma$ to obtain an all- n equivalent form.

transformations Each integer is a valuation/factor-incidence holon; $\sigma(n)$ is an aggregate divisor receiver which has compressed the individual paths.

boundary OPEN: prove the universal inequality. Checking integers, even extremely far, cannot exclude an arbitrarily late violation without an exact reduction.

source J. Lagarias, An Elementary Problem Equivalent to the Riemann Hypothesis .

13.0.16 RH.0150 Bagchi strong recurrence

Equivalence · proved-standard

depends: H.0019, H.0250, H.0254, H.0309, RH.0000

statement On the strip $D = \{s : \frac{1}{2} < \text{Re } s < 1\}$, RH is equivalent to positive-density strong self-recurrence: for every compact $K \subset D$ with connected complement and every $\varepsilon > 0$,

$$\liminf_{T \rightarrow \infty} \frac{1}{T} \text{meas} \left\{ \tau \in [0, T] : \sup_{s \in K} |\zeta(s + i\tau) - \zeta(s)| < \varepsilon \right\} > 0.$$

derivation Voronin-type universality supplies arbitrarily close neighborhood returns for nonvanishing target functions. Under RH, ζ is nonzero on D , so every declared neighborhood of its compact restriction recurs. Conversely, this exact recurrence property combined with Rouché/Hurwitz zero stability would reproduce any zero under vertical shifts; positive density is incompatible with the known zero geometry unless D is zero-free.

transformations This is exact self-similarity as recurrence after transport, not identity of occurrences. Vertical translation changes the occurrence while one compact receiver face enters every declared uniform neighborhood.

boundary OPEN: establish the positive-density recurrence for every compact receiver. One numerical recurrence or one compact does not close it.

source Bagchi’s strong-recurrence theorem; original dissertation record: Statistical Behaviour and Universality Properties of the Riemann Zeta Function and Other Allied Dirichlet Series .

13.0.17 RH.0160 RH falsity is finitely witnessed; independence is a separate metatheorem

Boundary · proved-derived

depends: H.0401, H.0403, H.0419, RH.0000, RH.0140

statement Exact arithmetic criteria such as RH.0140 give a decidable predicate $R_{\text{RH}(n)}$ for which

$$\text{RH} \leftrightarrow \forall n \in \mathbb{N}, R_{\text{RH}(n)}.$$

Therefore an exact machine M_{RH} may enumerate n , verify $R_{\text{RH}(n)}$, and halt at the first failure:

$$\neg \text{RH} \leftrightarrow \exists n, \neg R_{\text{RH}(n)} \leftrightarrow M_{\text{RH}} \text{ halts.}$$

A disproof consequently has a finite checkable witness. A proof must carry the universal quantifier by an invariant, descent, spectral theorem, positivity theorem in a declared orientation, or another finite proof object; it need not enumerate every integer or zero.

transformations In holonic language, the counterexample machine grows an exact causal current and stops only at a witnessed obstruction. Proving RH means proving that every admitted current preserves a law excluding that halt state. Proving independence from a theory T instead concerns the topology of T 's proof relation: one must establish both T does not prove RH and T does not prove $\neg$ RH.

boundary As of this registry, there is no accepted proof of RH, counterexample to RH, or proof that RH is independent of PA, ZFC, or another named standard foundation. Gödel incompleteness alone supplies none of those conclusions. Infinite extent, late-emerging structure, fractal self-similarity, and computational complexity explain why finite-prefix evidence is insufficient; they do not establish formal independence.

source Clay Mathematics Institute, current RH status Lagarias, an elementary criterion equivalent to RH Yedidia and Aaronson, explicit Turing machines for open statements Gödel incompleteness.

13.0.18 RH.0900 Four nonidentical proof mechanisms

Atlas · proved-derived

depends: RH.0010, RH.0020, RH.0030, RH.0040, RH.0050, RH.0060, RH.0070, RH.0080, RH.0090, RH.0100, RH.0110, RH.0120, RH.0130, RH.0140, RH.0150, RH.0160

statement The routes separate into four mechanism classes:

1. source-derived positive response: Weil, Pick, Stieltjes, de Branges;
2. exact closure and cancellation: Nyman–Beurling, prime/Möbius residual, Bagchi;
3. real-zero-preserving phase dynamics: Jensen, de Bruijn–Newman, Speiser;
4. exact source–spectrum geometry: Hilbert–Pólya, trace formula, cohomology.

transformations These are nonidentical receivers of the same completed arithmetic–analytic relation. A holonic contribution may construct natural transformations between them only when it proves that the route's proof-bearing property is preserved.

boundary No route presently closes its universal source-side obligation in this registry. Moving an obstruction to another receiver is useful only when the transition reduces it, supplies a new invariant, or yields a counterexample.

source Independent laboratory route audit against the cited primary sources.

14 Counterexamples and quarantined false bridges

14.0.1 C.0001 Contraction does not pass through arbitrary analysis and synthesis maps

Counterexample · counterexample

depends: RH.0010

statement The implication

$$K^*K \leq I \wedge CE = I \rightarrow (CKE)^*(CKE) \leq I$$

is false.

derivation Take

$$C = \begin{pmatrix} 1 & 0 \end{pmatrix}, \quad E = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad K = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Then $CE = 1$ and $K^*K = I$, but $CKE = 2$, so $(CKE)^*(CKE) = 4 > 1$.

transformations A valid lift needs

$$K^*C^*CK \leq C^*C$$

on the relevant range, or directly

$$E^*K^*C^*CKE \leq I.$$

boundary This rejects only the unsupported pull-through implication. It does not reject a route that separately proves the weighted inequality.

source Exact laboratory counterexample to a recent Volterra/Weyl lift.

14.0.2 C.0002 Stationarity on a constraint fiber does not imply positivity

Counterexample · counterexample

depends: H.0013

statement Nonnegativity on a constraint kernel together with an orthogonal or stationary chosen representative need not make the full quadratic form nonnegative.

derivation Let

$$Q(x, y) = y^2 - 3x^2, \quad R(x, y) = x.$$

On $\ker R = \{x = 0\}$, $Q(0, y) = y^2 \geq 0$. The representative $(x, 0)$ is stationary/orthogonal in the fiber direction, yet $Q(x, 0) = -3x^2 < 0$ for $x \neq 0$.

boundary Euler–Lagrange orthogonality supplies a criticality condition, not the missing global sign.

source Exact laboratory counterexample.

14.0.3 C.0003 Finite verification cannot discharge an unbounded criterion

Counterexample · counterexample

depends: RH.0030, RH.0060, RH.0140

statement From $P(n)$ for all $n \leq N$ one cannot infer $\forall n \in \mathbb{N} : P(n)$ without an induction, monotonicity, finite reduction, or other theorem connecting the tail.

derivation For any chosen N , define $P(n)$ to be $n \leq N$. Every checked case succeeds and $P(N + 1)$ fails.

boundary This does not diminish computational witnesses. It fixes their grade and identifies the exact bridge an infinite claim still owes.

source Elementary logical counterexample.

14.0.4 C.0004 Positive source moments need not survive a nonlinear logarithmic transform

Counterexample · counterexample

depends: RH.0040

statement Positivity of the coefficients or moments of a source function does not by itself imply that the coefficients of its logarithmic derivative form a Stieltjes moment sequence.

derivation The polynomial $A(w) = 1 + w + w^2$ has positive coefficients, but its zeros are the nonreal primitive cube roots of unity. Hence $\frac{A'}{A}$ has nonreal poles and cannot be a Stieltjes transform of a positive measure supported on $[0, \infty)$ after the required negative-axis convention.

boundary The theta source may possess additional total-positivity or variation structure; that structure must be proved to survive the logarithmic transform rather than inferred from coefficient positivity.

source Elementary polynomial counterexample.

14.0.5 C.0005 Pointwise convergence does not imply Hilbert-space closure

Counterexample · counterexample

depends: RH.0050

statement Pointwise convergence of f_n to f does not imply $\|f_n - f\|_2 \rightarrow 0$.

derivation On $(0, 1)$ let

$$f_{n(x)} = \sqrt{n} 1_{(0, \frac{1}{n})}(x).$$

For every fixed $x > 0$, $f_{n(x)} \rightarrow 0$, but

$$\|f_n\|_2^2 = n \left(\frac{1}{n} \right) = 1$$

for all n .

boundary A Nyman–Beurling finite-span sequence must converge in its declared L^2 norm, not merely at sampled points.

source Standard analysis counterexample.

14.0.6 C.0006 The historical natural-square-root surrogate is not RH

Boundary · historical-toy

depends: RH.0000

statement The old proposition

$$\forall n \in \mathbb{N} : s(n)^2 \leq n < (s(n) + 1)^2$$

for an integer square-root algorithm is true by construction but has no interpretation map to the nontrivial zeros of ζ .

derivation It quantifies over natural numbers and the correctness of integer square root. RH.0000 quantifies over complex zeros of the analytically continued completed zeta function. No map carrying zeros, real parts, or the functional equation into the surrogate was proved.

boundary The old Lean file may remain historical testimony. It cannot be imported as an RH theorem or holonic soundness result.

source `src/labyrinth/mathematics/lean/App_RH.lean`.

14.0.7 C.0007 An inhabited OPEN sentinel proves no open conjecture

Boundary · historical-toy

depends: H.0001

statement If

inductive OPEN where | stillOpen : OPEN,

then a theorem returning OPEN.stillOpen proves only that the custom type OPEN is inhabited. It does not prove or even state Goldbach, RH, Navier–Stokes, or another named conjecture.

boundary An open conjecture must be encoded as its actual proposition with no supplied inhabitant. Conditional results accept it as an explicit hypothesis.

source Audit of the historical Labyrinth Lean corpus.

14.0.8 C.0008 A digest is neither identity nor proof

Counterexample · counterexample

depends: H.0015

statement A hash $h : X \rightarrow B$ can locate or compare serialized faces. Unless injectivity on the declared domain is proved, $h(x) = h(y)$ does not imply $x = y$; even injectivity of bytes would not prove equality of causal diagrams represented by those bytes.

derivation A finite digest codomain and a larger finite or infinite source domain force collisions by the pigeonhole principle. Separately, one byte string may be decoded by nonidentical semantics.

boundary Digests remain useful provenance and certificate locators. They do not replace independent checking or soul-equivalence.

source Elementary cardinality and semantic-boundary argument.